

# CNS Mid MCQs

2026

Answered



+ 1000 MCQs

Medico Helpers

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# Scalp, face and muscles of the head

**1- Infection can spread from the scalp to the cranial cavity through:**

- A- Arteries
- B- Emissary veins**
- C- Lymphatics
- D- Nerves

**2- The occipitofrontalis muscle frontalis belly arises from:**

- A- Zygomatic arch
- B- Mandible
- C- Skin of eyebrows
- D- Front of the epicranial aponeurosis**

**3- The occipital belly of occipitofrontalis is supplied by:**

- A- Temporal nerve
- B- Posterior auricular nerve**
- C- Zygomatic nerve
- D- Cervical nerve

**4- The scalp is supplied anteriorly by the:**

- A- Occipital artery
- B- Posterior auricular artery
- C- Supratrochlear artery**
- D- Infraorbital artery

**5- Which is not a branch of external carotid artery?**

- A- Superficial temporal artery
- B- Supraorbital artery**
- C- Posterior auricular artery
- D- Occipital artery

**6- The auriculotemporal nerve supplies which area?**

- A- Forehead
- B- Back of scalp
- C- Lateral scalp**
- D- Occipital region

**7- Which is Not supply scalp behind the ear**

- A- 3rd occipital nerve
- B- Great auricular nerves
- C- Lesser occipital nerve
- D- Auriculotemporal nerve**

**8- The supratrochlear and supraorbital nerves are branches of:**

- A- CN V
- B- CN V3
- C- CN V1**
- D- CN V2

**9- Orbicularis oculi muscle innervated by**

- A- Marginal mandibular nerve
- B- Posterior auricular nerve
- C- Zygomatic nerve**
- D- Buccal nerve

**10- Parotid duct and buccal branch of mandibular nerve pierce which muscle**

- A- Orbicularis oris muscle
- B- Orbicularis oculi muscle
- C- Buccinator muscle**

**11- The facial nerve exits the skull through:**

- A- Foramen spinosum
- B- Foramen lacerum
- C- Stylomastoid foramen**
- D- Jugular foramen

**12- In Bell's palsy, inability to elevate eyebrow is due to paralysis of**

- A- Orbicularis oculi muscle
- B- Frontalis muscle**
- C- Orbicularis oris muscle
- D- Buccinator muscle

**13- Deviation of mouth to the normal side occurs because:**

- A- Affected side muscles are hyperactive
- B- Trigeminal nerve palsy
- C- Normal side muscles pull the mouth laterally**
- D- Both sides contract equally

**14- Drooling of saliva in Bell's palsy occurs because of paralysis of:**

- A- Frontalis
- B- Buccinator & orbicularis oris**
- C- Temporalis
- D- Occipitalis

**15- Sensory supply of the face is mainly from:**

- A- Facial nerve
- B- Trigeminal nerve**
- C- Glossopharyngeal nerve
- D- Hypoglossal nerve

**16- The mental nerve is a branch of:**

- A- Ophthalmic nerve
- B- Maxillary nerve
- C- Mandibular nerve**
- D- Facial nerve

**17- The zygomaticofacial nerve supplies:**

- A- Chin
- B- Lower lip
- C- Prominence of cheek**
- D- Nose bridge

**18- Skin over the angle of the mandible innervated by**

- A- Facial nerve
- B- Mandibular nerve
- C- Trigeminal nerve
- D- Great auricular nerve**

**19- Trigeminal sensory loss over the forehead indicates lesion of:**

- A- Mandibular nerve
- B- Maxillary nerve
- C- Ophthalmic nerve**
- D- Facial nerve

**20- All muscles of mastication insert on:**

- A- Maxilla
- B- Mandible**
- C- Temporal bone
- D- Zygomatic arch

**21- Which muscle not elevates the mandible?**

- A- Medial pterygoid
- B- Masseter
- C- Temporalis
- D- Lateral pterygoid**

**22- The muscle that helps in opening the mouth actively is:**

- A- Medial pterygoid
- B- Lateral pterygoid**
- C- Masseter
- D- Temporalis

**23- Temporalis muscle inserted in which part of mandible**

- A- Condylod process
- B- Angle and ramus
- C- Coronoid process**
- D- Mental protuberance

**24- The primary arterial supply to the face is from branches of:**

- A- Internal carotid artery
- B- External carotid artery**
- C- Subclavian artery
- D- Vertebral artery

**25- The mental artery is a direct continuation of the:**

- A- Facial artery
- B- Infraorbital artery
- C- Inferior alveolar artery**
- D- Superficial temporal artery

**26- The dangerous area of the face drains into the cavernous sinus via:**

- A- Retromandibular vein
- B- Deep temporal vein
- C- Facial vein and ophthalmic veins**
- D- External jugular vein

**27- The TMJ articular surface of temporal bone is covered by:**

- A- Hyaline cartilage
- B- Fibrocartilage**
- C- Elastic cartilage
- D- No cartilage

**28- The joint cavity of TMJ is divided into two compartments by:**

- A- Capsule
- B- Articular disc**
- C- Meniscus
- D- Ligament

**29- Hinge movement (depression/elevation) occurs in the:**

- A- Upper compartment
- B- Lower compartment**
- C- Middle compartment
- D- Outside capsule

**30- The stylomandibular ligament runs from:**

- A- Styloid process to mandible**
- B- Sphenoid to mandible
- C- Styloid to temporal bone
- D- Styloid to occipital bone

**31- TMJ is mainly innervated by:**

- A- Facial nerve
- B- Glossopharyngeal nerve
- C- Mandibular divisions nerve**
- D- Hypoglossal nerve

**32- Muscles of tongue are supplied by:**

- A- Facial nerve
- B- Trigeminal nerve
- C- Hypoglossal nerve**
- D- Glossopharyngeal

**33- Which muscle originated from greater horn of the hyoid bone**

- A- Genioglossus
- B- Styloglossus
- C- Palatoglossus
- D- Hyoglossus**

**34- Which muscle not insert in side of the tongue**

- A- Hyoglossus
- B- Styloglossus
- C- Palatoglossus
- D- Genioglossus**

**35- Constrictors muscles have a Common origin from Median pharyngeal raphe**

- A- ☒
- B- ☐**

**36- Stylopharyngeus muscle supplied by**

- A- Hypoglossal nerve
- B- Glossopharyngeal nerve**
- C- Cranial accessory through vagus
- D- Facial nerve

**37- Palatoglossus muscle action on the the tongue**

- A- Depress
- B- Protrude
- C- Raise**
- D- Retract

**38- Which muscle retract and elevate the tongue**

- A- Genioglossus
- B- Hyoglossus
- C- Styloglossus**
- D- Palatoglossus

**39- Which muscle not supplied by cranial root of accessory nerve**

- A- Levator palati
- B- Palatopharyngeus
- C- Inferior constrictor
- D- Tensor palati**

**40- Which muscle elevates the larynx and pharynx and depresses the soft palate**

- A- Levator palati
- B- Palatoglossus
- C- Palatopharyngeus**
- D- Tensor palate

**41- Which Right about Tensor plate muscle**

- A- Originate from quadrants area on the petrous temporal bone
- B- Supply by cranial root of accessory nerve
- C- Tense the palatine aponeurosis and opens auditory tube**
- D- Insert in side of the tongue

**42- The angular artery is the terminal branch of the:**

- A- Maxillary artery
- B- Lingual artery
- C- Facial artery**
- D- Superficial temporal artery

**43- The retromandibular vein passes inferiorly through the:**

- A- Submandibular space
- B- Parotid gland**
- C- Buccinator muscle
- D- Nasopharynx

**44- The common facial vein drains into the:**

- A- External jugular vein
- B- Internal jugular vein**
- C- Subclavian vein
- D- Vertebral vein

**Formatives****& Previous Exams****45- Auriculo-temporal nerve is a branch of:**

- A- Maxillary nerve
- B- Mandibular nerve**
- C- Ophthalmic nerve
- D- Facial nerve

**46- Infraorbital nerve is a continuation of:**

- A- Maxillary nerve**
- B- Mandibular nerve
- C- Ophthalmic nerve
- D- Facial nerve

**47- When the muscle contracts, it makes tongue depression:**

- A- Mylo-hyoid muscle
- B- Digastric muscle
- C- Hyo-glossus muscle**
- D- Stylo-glossus muscle

**48- The main action of buccinator muscle is:**

- A- Eye closure
- B- Mouth closure
- C- Prevent food accumulation in the mouth vestibule**
- D- Mandible depression

**49- The retro-mandibular vein is formed by union of:**

- A- Maxillary & superficial temporal veins**
- B- Supra-orbital & supra-trochlear veins
- C- External & internal jugular veins
- D- Internal jugular & subclavian veins

**50- The dangerous layer of the scalp is:**

- A- Skin layer
- B- Subcutaneous connective tissue layer
- C- Epicranial aponeurosis
- D- Loose connective tissue layer**

**51- The mastication muscle which is concerned with mandibular retraction is:**

- A- Masseter
- B- Anterior fibers of temporalis
- C- Posterior fibers of temporalis**
- D- Medial & lateral pterygoids

**52- The strongest ligament of T/M joint is:**

- A- Pterygo-mandibular ligament
- B- Spheno-mandibular ligament
- C- Stylo-mandibular ligament
- D- Lateral ligament of T/M joint**

**53- Which vein is responsible for draining the majority of the external face?**

- A- External jugular vein**
- B- Anterior jugular vein
- C- Internal jugular vein
- D- Posterior auricular vein

**54- What is the most internal layer of the scalp?**

- A- Epicranial aponeurosis
- B- Periosteum**
- C- Dense connective tissue
- D- Dura mater

**55- Contraction of left lateral pterygoid will result in:**

- A- Deviation of mandible upwards
- B- Deviation of mandible downwards
- C- Deviation of mandible to right side**
- D- Deviation of mandible to left side

**56- The skin area over the angle of the mandible is supplied by**

- A- Supra-orbital nerve
- B- Great auricular nerve**
- C- Mandibular nerve
- D- Maxillary nerve

**57- The nerve supply to the posterior belly of digastrics is**

- A- Facial nerve**
- B- Mandibular nerve
- C- Maxillary nerve
- D- Ophthalmic nerve

**58- The common facial vein ends into:**

- A- Internal jugular vein**
- B- Anterior jugular vein
- C- External jugular vein
- D- Posterior jugular vein

**59- Which is an intrinsic muscle of the tongue**

- A- Superior longitudinal ms**
- B- Genioglossus
- C- Hyoglossus
- D- Styloglossus

**60- External jugular vein terminates by draining into**

- A- Post auricular vein
- B- Retromandibular vein (post branche)
- C- Retromandibular vein (ant branche)
- D- Subcalvian vein**

**61- The medial pretrygoid follows the same angle as masseter arising from medial pterygoid plate, where does it attach on the mandible**

- A- Ramus and angle**
- B- Condylod process
- C- Coronoid process
- D- Mantel protuberance

**62- The deep structures of face are supplied by**

- A- Superficial temporal art
- B- Facial art
- C- Subclavian art
- D- Maxillary art**

**63- Damage to the hypoglossal nerve on one side causes the tongue to deviate toward the affected side because of paralysis of which muscle?**

- A- Hyoglossus
- B- Styloglossus
- C- Genioglossus
- D- Palatoglossu

**64- All muscles of the soft palate are supplied by the cranial root of the accessory nerve via the vagus EXCEPT**

- A- Levator palati
- B- Palatopharyngeus
- C- Palatoglossus
- D- Tensor palate

**65- The "danger area of the face" (around nose and upper lip) is clinically important because infection can spread to the cavernous sinus through which venous pathway?**

- A- Internal jugular vein
- B- External jugular vein
- C- Ophthalmic veins
- D- Retromandibular vein

**66- Which muscle is responsible for protrusion of the mandible and opening of the mouth?**

- A- Masseter
- B- Temporalis
- C- Medial pterygoid
- D- Lateral pterygoid

# Muscles of the neck

**1- Which muscle forms the posterior boundary of the anterior triangle of the neck?**

- A- Trapezius muscle
- B- Posterior belly of the digastric
- C- Anterior border of the sternocleidomastoid muscle**
- D- Superior belly of the omohyoid

**2- What are the primary contents of the submental triangle**

- A- Common carotid artery and internal jugular vein
- B- Hypoglossal nerve and facial artery
- C- Submental lymph nodes and submental veins**
- D- Strap muscles and thyroid gland

**3- The inferior belly of the omohyoid muscle divides the posterior triangle of the neck into which two sub-triangles?**

- A- Digastric and Carotid triangles
- B- Muscular and Submental triangles
- C- Occipital and subclavian triangles**
- D- Superior and Inferior triangles

**4- Which nerve supplies the posterior belly of the digastric muscle?**

- A- Mandibular nerve (V3) (CN XII)
- B- Facial nerve (CN VII)**
- C- Hypoglossal nerve
- D- Ansa cervicalis

**5- All infra-hyoid (strap) muscles are supplied by the ansa cervicalis EXCEPT for which muscle?**

- A- Sternohyoid muscle
- B- Sternothyroid muscle
- C- Omohyoid muscle
- D- Thyrohyoid muscle**

**6- Which muscle acts to elevate the hyoid bone during swallowing and can also depress the mandible when the hyoid bone is fixed?**

- A- Sternohyoid muscle
- B- Digastric muscle
- C- Mylohyoid muscle**
- D- Thyrohyoid muscle

**7- Which of the following structures is NOT a content of the Carotid triangle?**

- A- Common carotid artery
- B- Hypoglossal nerve
- C- Nerve to mylohyoid**
- D- External carotid artery

**8- The simultaneous contraction of both the sternocleidomastoid muscles results in which action?**

- A- Neck rotation to the opposite side
- B- Lateral flexion of the neck to the same side
- C- Neck flexion**
- D- Elevation of the first and second ribs

**9- The middle scalene muscle is inserted into which structure?**

- A- Scalene tubercle of the first rib
- B- Second rib
- C- Posterior surface of the manubrium
- D- First rib, posterior to the insertion of the anterior scalene muscle**

**10- Which muscle is anatomically described as forming the floor of the oral cavity (oral diaphragm)?**

- A- Platysma
- B- Stylohyoid muscle
- C- Mylohyoid muscle**
- D- Geniohyoid muscle

**11- The hypoglossal nerve (CN XII) is a key content of which sub-division of the anterior triangle?**

- A- Carotid triangle
- B- Muscular triangle
- C- Submental triangle
- D- Digastric (submandibular) triangle**

**12- Which extrinsic muscle of the tongue is the exception to the Hypoglossal nerve supply?**

- A- Genioglossus
- B- Hyoglossus
- C- Styloglossus
- D- Palatoglossus**

**13- The Cervical nerve, a terminal branch of the facial nerve, supplies which muscle?**

- A- Occipitalis
- B- Stylohyoid
- C- Buccinator
- D- Platysma**

**14- When swallowing (degultition), the strap muscles of the neck perform which essential action?**

- A- Elevate the hyoid bone
- B- Depress the hyoid bone and the thyroid cartilage**
- C- Elevate the pharynx and larynx
- D- Protrude the mandible

**15- The digastric muscle is unique because its two bellies have different nerve supplies Which of the following correctly describes the innervation of the posterior belly?**

- A- Mandibular nerve
- B- Hypoglossal nerve
- C- Facial nerve**
- D- Ansa cervicalis

**16- During the process of deglutition what is the primary function of the infra-hyoid "strap" muscles?**

- A- Elevating the hyoid bone
- B- Depressing the hyoid bone and thyroid cartilag**
- C- Rotating the face to the opposite side
- D- Elevating the first and second rib

**17- Which anatomical structure forms the inferior boundary (base) of the posterior triangle of the neck?**

- A- Midline of the neck
- B- Trapezius muscle
- C- Sternocleidomastoid muscle
- D- Clavicle**

**18- The scalene muscles are considered accessory muscles of respiration. How do they contribute to the process of inspiration?**

- A- By depressing the hyoid bone
- B- By elevating the first and second ribs**
- C- By flexing the neck forward on the chest
- D- By narrowing the floor of the oral

**19- The posterior triangle of the neck is divided into the occipital and subclavian triangles by which specific structure?**

- A- Superior belly of the omohyoid muscle
- B- Inferior belly of the omohyoid muscle**
- C- Middle scalene muscle
- D- Sternal head of the sternocleidomastoid muscle

**20- While most infra-hyoid (strap) muscles are supplied by the ansa cervicalis, which muscle is an exception and is supplied by C1 fibres via the hypoglossal nerve?**

- A- Sternohyoid
- B- Sternothyroid
- C- Thyrohyoid**
- D- Omohyoid

**21- Under what condition does the mylohyoid muscle act to depress the mandible?**

- A- When the hyoid bone is fixed
- B- During the initial stage of swallowing
- C- When the neck is flexed
- D- When the facial nerve is stimulated

**22- The muscular triangle contains the strap muscles of the neck, which structure do these muscles cover?**

- A- Submandibular gland
- B- Parotid gland
- C- Thyroid gland
- D- Brachial plexus

**23- A patient is unable to depress their mandible while their hyoid bone is fixed, which nerve is likely damaged, and which other muscle would also be paralysed as a result of this specific nerve injury?**

- A- Facial nerve; Posterior belly of the digastric
- B- Mandibular nerve; Anterior belly of the digastric
- C- Ansa cervicalis; Sternohyoid
- D- Hypoglossal nerve; Thyrohyoid

**24- A surgeon performing an operation in the anterior triangle of the neck needs to locate the common carotid artery in which specific subdivision of the anterior triangle is this artery located?**

- A- Digastric triangle
- B- Carotid triangle
- C- Muscular triangle
- D- Submental triangle

## Formatives

### & Previous Exams

**25- The nerve supply of the mylohyoid muscle is**

- A- Facial nerve
- B- Mandibular nerve
- C- Maxillary nerve
- D- Ophthalmic nerve

**26- Subclavian vessels are contents of**

- A- Submental triangle.
- B- Digastric triangle
- C- Posterior triangle
- D- Muscular triangle.

**27- Phrenic nerve is present in the neck in front of:**

- A- Scalenus anterior muscle
- B- Scalenus posterior muscle
- C- Mylo-hyoid muscle
- D- Sterno-mastoid muscles

**28- When the muscle contracts, it makes tongue depression:**

- A- Mylo-hyoid muscle.
- B- Digastric muscle
- C- Hyo-glossus muscle.
- D- Stylo-glossus muscle.

**29- Which of the following infrahyoid muscles are found in deep plane**

- A- Sternohyoid and thyrohyoid
- B- Sternohyoid and omohyoid
- C- Sternohyoid and sternohyoid
- D- Sternohyoid and thyrohyoi

**30- Which of the following cranial nerves can be found in the anterior triangle?**

- A- Oculomotor (CN III)
- B- Vestibulococlear
- C- Trochlear
- D- vagus

**31- The nerve supply to the posterior belly of digastrics is**

- A- Facial nerve.
- B- Mandibular nerve
- C- Maxillary nerve
- D- Ophthalmic nerve.

**32- Regarding the triangles of the neck**

- A- Post triangle is bound by trapezius, sternomastoid and the mandible
- B- Ant triangle is divided into two triangles
- C- Post belly of digastric separates the carotid and submandibular triangle
- D- Accessory N divides the post triangle into lower small and upper large triangle

**33- Carotid triangle contains**

- A- Glossopharyngeal N
- B- Hypoglossal N
- C- Facial N
- D- Mandibular N

**34- Which two muscles, covering the thyroid gland, are listed as the contents of the Muscular Triangle?**

- A- Digastric and Stylohyoid muscles
- B- Omohyoid and Sternocleidomastoid muscle
- C- Sterno-hyoid and Sterno-thyroid muscles
- D- Scalene and Trapezius muscles

**35- The Mylohyoid muscle, which forms the floor of the oral cavity (oral diaphragm), receives its innervation from which nerve?**

- A- Facial nerve
- B- Hypoglossal nerve
- C- Mandibular nerve
- D- Vagus nerve

# Nerves of head & neck I, II

**1- The olfactory nerve is classified as.....?**

- A- Special visceral efferent
- B- Special somatic afferent**
- C- General visceral efferent
- D- General somatic afferent

**2- Olfactory nerve fibers are formed by axons of.....?**

- A- Ganglion cells
- B- Olfactory receptor cells**
- C- Mitral cells
- D- Bipolar cells

**3- Olfactory nerve fibers pass through the.....?**

- A- Superior orbital fissure
- B- Jugular foramen
- C- Cribriform plate**
- D- Optic canal

**4- Olfactory nerve terminates in the.....?**

- A- Thalamus
- B- Piriform cortex
- C- Olfactory bulb**
- D- Olfactory tract

**5- Olfactory mucosa is present in the....?**

- A- Oropharynx
- B- Upper part of nasal cavity**
- C- Nasopharynx
- D- Lower nasal cavity

**6- Optic nerve is a.....?**

- A- Autonomic nerve
- B- Special somatic afferent nerve**
- C- Mixed nerve
- D- Motor nerve

**7- Optic nerve fibers arise from.....?**

- A- Amacrine cells
- B- Ganglion cells of retina**
- C- Bipolar cells
- D- Rods and cones

**8- The most anterior part of optic nerve is the.....?**

- A- Intracranial part
- B- Intraocular part**
- C- Intracanalicular part
- D- Intraorbital part

**9- Optic nerve passes through the....?**

- A- Foramen rotundum
- B- Optic canal**
- C- Inferior orbital fissure
- D- Superior orbital fissure

**10- Nasal retinal fibers.....?**

- A- End in ipsilateral optic tract
- B- Cross in optic chiasma**
- C- Cross in optic canal
- D- Do not cross

**11- Temporal retinal fibers....?**

- A- End in superior colliculus
- B- Do not cross**
- C- End in contralateral optic tract
- D- Cross in optic chiasma

**12- Optic tract fibers relay in the.....?**

- A- Visual cortex
- B- Hypothalamus
- C- Lateral geniculate body**
- D- Superior colliculus

**13- Primary visual cortex is located in the.....?**

- A- Temporal lobe
- B- Lips of calcarine sulcus**
- C- Parietal lobe
- D- Cingulate gyrus

**14- Which of the following is the correct order of Optic nerve pathway.....?**

- A- Intraorbital → Intraocular → Intracanalicular → Intracranial
- B- Intraocular → Intraorbital → Intracanalicular → Intracranial**
- C- Intraorbital → Intracanalicular → Intraocular → Intracranial
- D- Intraocular → Intraorbital → Intracranial → Intracanalicular

**15- Oculomotor nerve is.....?**

- A- Pure motor**
- B- Pure sensory
- C- Mixed (sensory, motor)
- D- Autonomic only

**16- Oculomotor nerve originates from....?**

- A- Thalamus
- B- Medulla
- C- Upper midbrain**
- D- Lower pons

**17- Parasympathetic nucleus of CN III is.....?**

- A- Inferior salivatory nucleus
- B- Solitary nucleus
- C- Edinger–Westphal nucleus**
- D- Red nucleus

**18- CN III enters the orbit through....?**

- A- Foramen rotundum
- B- Superior orbital fissure**
- C- Inferior orbital fissure
- D- Optic canal

**19- Oculomotor nerve divides into.....?**

- A- Anterior and posterior divisions
- B- Superior and inferior divisions**
- C- Medial and lateral divisions
- D- No divisions

**20- CN III supplies all extraocular muscles except.....?**

- A- Inferior oblique
- B- Lateral rectus**
- C- Medial rectus
- D- Inferior rectus

**21- Parasympathetic fibers of CN III supply.....?**

- A- Superior rectus
- B- Lacrimal gland
- C- Constrictor pupillae**
- D- Dilator pupillae

**22- CN III lesion causes.....?**

- A- Vertical squint only
- B- No squint
- C- Lateral squint**
- D- Medial squint

**23- Ptosis in CN III lesion is due to paralysis of.....?**

- A- Superior oblique
- B- Frontalis
- C- Levator palpebrae superioris**
- D- Orbicularis oculi

**24- Pupillary dilation in CN III lesion is due to loss of.....?**

- A- Sensory fibers
- B- Motor fibers
- C- Parasympathetic supply**
- D- Sympathetic supply

**25- Trochlear nerve supplies.....?**

- A- Lateral rectus
- B- Superior oblique**
- C- Superior rectus
- D- Inferior oblique

**26- Trochlear nerve enters orbit through.....?**

- A- Foramen ovale
- B- Superior orbital fissure**
- C- Inferior orbital fissure
- D- Optic canal

**27- Trochlear nerve lesion causes.....?**

- A- Lateral squint
- B- Medial squint
- C- Diplopia**
- D- Ptosis

**28- Abducent nerve supplies.....?**

- A- Superior rectus
- B- Lateral rectus**
- C- Inferior rectus
- D- Medial rectus

**29- Abducent nerve lesion causes.....?**

- A- Pupil dilation
- B- Ptosis
- C- Medial squint**
- D- Lateral squint

**30- Abducent nerve enters orbit through.....?**

- A- Foramen ovale
- B- Superior orbital fissure**
- C- Inferior orbital fissure
- D- Optic canal

**31- Abducent nerve originates from.....?**

- A- Motor nucleus of MO
- B- Motor nucleus of lower pons**
- C- Motor nucleus of upper pons
- D- Motor nucleus of midbrain

**32- Trigeminal nerve is.....?**

- A- Autonomic nerve
- B- Mixed nerve**
- C- Sensory only
- D- Motor only

**33- Trigeminal nerve attaches to pons by.....?**

- A- Four roots
- B- Three roots
- C- Two roots**
- D- One root

**34- Trigeminal nerve has.....?**

- A- Four sensory nuclei
- B- Three sensory nuclei**
- C- Two sensory nuclei
- D- One sensory nucleus

**35- Trigeminal ganglion is located at.....?**

- A- Foramen ovale
- B- Jugular foramen
- C- Apex of petrous bone**
- D- Cavernous sinus

**36- Ophthalmic nerve is.....?**

- A- Parasympathetic
- B- Pure sensory**
- C- Pure motor
- D- Mixed

**37- V1 passes through.....?**

- A- Inferior orbital fissure
- B- Superior orbital fissure**
- C- Foramen ovale
- D- Foramen rotundum

**38- Frontal nerve divides into.....?**

- A- Ethmoidal branches
- B- Zygomatic branches
- C- Infraorbital, lacrimal
- D- Supraorbital, supratrochlear**

**39- Lacrimal nerve supplies....?**

- A- Eyeball muscles
- B- Retina
- C- Cornea
- D- Lacrimal gland**

**40- Naso-ciliary nerve gives.....?**

- A- Auriculotemporal nerve
- B- Chorda tympani
- C- Greater petrosal nerve
- D- Long ciliary nerves**

**41- Long ciliary nerves supply.....?**

- A- Optic disc
- B- Lacrimal gland
- C- Cornea, iris, ciliary body**
- D- Retina

**42- Infratrochlear nerve supplies.... ?**

- A- Upper lip
- B- Cheek
- C- Medial eyelids**
- D- Forehead

**43- Maxillary nerve is.....?**

- A- Parasympathetic
- B- Motor
- C- Sensory**
- D- Mixed

**44- V2 exits skull via.....?**

- A- Optic canal
- B- Foramen rotundum**
- C- Foramen spinosum
- D- Foramen ovale

**45- PSA nerve supplies.....?**

- A- Molars**
- B- Premolars
- C- Canines
- D- Incisors

**46- Middle superior alveolar nerve supplies.....?**

- A- Molars
- B- Premolars**
- C- Canines
- D- Incisors

**47- Infraorbital nerve appears on face through.....?**

- A- Zygomatic foramen
- B- Supraorbital foramen
- C- Mental foramen
- D- Infraorbital foramen**

**48- Mandibular nerve is.....?**

- A- Parasympathetic
- B- Mixed**
- C- Motor
- D- Sensory

**49- V3 exits skull via.....?**

- A- Jugular foramen
- B- Foramen spinosum
- C- Foramen ovale**
- D- Foramen rotundum

**50- Nervus spinosus supplies.....?**

- A- Parotid gland
- B- Dura mater**
- C- Middle ear
- D- TMJ

**51- Buccal nerve is.....?**

- A- Mixed
- B- Sensory**
- C- Parasympathetic
- D- Motor

**52- Lingual nerve carries taste from....?**

- A- Soft palate
- B- Epiglottis
- C- Anterior 2/3 of tongue**
- D- Posterior 1/3 of tongue

**53- Inferior alveolar nerve supplies.....?**

- A- Cheek
- B- Tongue
- C- Lower teeth**
- D- Upper teeth

**54- Mental nerve supplies.....?**

- A- Lower lip, chin**
- B- Upper lip
- C- Nose
- D- Cheek

**55- Nerve to mylohyoid supplies.....?**

- A- Buccinator
- B- Masseter
- C- Mylohyoid, anterior belly of digastric**
- D- Tongue muscles

**56- Submandibular parasympathetic ganglion is suspended by.....?**

- A- Lesser petrosal nerve
- B- Lingual nerve**
- C- Chorda tympani nerve
- D- Oculomotor nerve

**57- Facial nerve is.....?**

- A- Autonomic only
- B- Mixed**
- C- Sensory only
- D- Motor only

**58- Facial nerve has all nuclei except....?**

- A- Inferior salivatory nucleus**
- B- Superior salivatory nucleus
- C- Solitary nucleus
- D- Motor nucleus

**59- Facial nerve exits skull through..... ?**

- A- Internal acoustic meatus
- B- Foramen ovale
- C- Jugular foramen
- D- Stylomastoid foramen**

**60- Greater petrosal nerve supplies.....?**

- A- Sublingual gland
- B- Submandibular gland
- C- Lacrimal gland**
- D- Parotid gland

**61- Chorda tympani carries....?**

- A- Sympathetic fibers
- B- Parasympathetic + taste**
- C- Sensory fibers
- D- Motor fibers

**62- Number of terminal branches of facial nerve....?**

- A- 6
- B- 5**
- C- 4
- D- 3

**63- Marginal mandibular nerve supplies muscles of.....?**

- A- Neck
- B- Eyelids
- C- Upper lip
- D- Lower lip**

**64- Submandibular ganglion is supplied by parasympathetic fibers of..... nerve?**

- A- Abducent
- B- Chorda tympani**
- C- Oculomotor
- D- Lesser petrosal

**65- Ciliary ganglion is supplied by parasympathetic fibers of.....?**

- A- Chorda tympani nerve
- B- Lesser petrosal nerve
- C- Greater petrosal nerve
- D- Oculomotor nerve**

**66- Vestibulocochlear nerve is.....?**

- A- Sensory**
- B- Motor
- C- Mixed
- D- Sympathetic

**67- CN VIII consists of.....?**

- A- Facial and cochlear nerves
- B- Vestibular and cochlear nerves**
- C- Vestibular nerve only
- D- Cochlear nerve only

**68- The cochlear division receives impulses from....?**

- A- Macula utriculi
- B- Organ of Corti**
- C- Vestibule
- D- Semicircular canals

**69- The vestibular division is concerned with....?**

- A- Vision
- B- Taste
- C- Balance and motion**
- D- Hearing only

**70- Vestibulocochlear nerve enters the cranial cavity through....?**

- A- Foramen magnum
- B- Stylomastoid foramen
- C- Internal acoustic meatus**
- D- Jugular foramen

**71- CN VIII attaches to the brainstem at the....?**

- A- Midbrain
- B- Lower border of pons**
- C- Medulla
- D- Upper pons

**72- Glossopharyngeal nerve is.....?**

- A- Sympathetic
- B- Mixed**
- C- Sensory only
- D- Motor only

**73- CN IX arises from all the following nuclei except.....?**

- A- Superior salivatory nucleus**
- B- Solitary nucleus
- C- Inferior salivatory nucleus
- D- Nucleus ambiguus

**74- Glossopharyngeal nerve exits the skull through.....?**

- A- Foramen rotundum
- B- Stylomastoid foramen
- C- Jugular foramen**
- D- Foramen ovale

**75- CN IX supplies taste sensation to....?**

- A- Soft palate
- B- Epiglottis
- C- Posterior 1/3 of tongue**
- D- Anterior 2/3 of tongue

**76- Stylopharyngeus muscle is supplied by.....?**

- A- Hypoglossal nerve
- B- Glossopharyngeal nerve**
- C- Vagus nerve
- D- Facial nerve

**77- Tympanic branch of CN IX carries parasympathetic fibers to.....?**

- A- Lacrimal gland
- B- Parotid gland**
- C- Sublingual gland
- D- Submandibular gland

**78- Carotid body is supplied by....?**

- A- Glossopharyngeal nerve**
- B- Facial nerve
- C- Olfactory nerve
- D- Oculomotor nerve

**79- CN IX enters the pharynx between....?**

- A- Palatoglossus, palatopharyngeus
- B- Superior, middle constrictors**
- C- Middle, inferior constrictors
- D- Superior, inferior constrictors

**80- Vagus nerve is.....?**

- A- Parasympathetic only
- B- Mixed**
- C- Sensory
- D- Motor

**81- Parasympathetic nucleus of vagus nerve is.....?**

- A- Solitary nucleus
- B- Dorsal motor nucleus of vagus**
- C- Superior salivatory nucleus
- D- Inferior salivatory nucleus

**82- Vagus nerve exits skull through...?**

- A- Foramen ovale
- B- Jugular foramen**
- C- Stylomastoid foramen
- D- Foramen magnum

**83- In the neck, vagus nerve lies in the carotid sheath between.....?**

- A- ECA, IJV
- B- ICA, IJV**
- C- ICA, ECA
- D- CCA, IJV

**84- Pharyngeal muscles supplied by vagus nerve except....?**

- A- Constrictors
- B- Palatopharyngeus
- C- Stylopharyngeus**
- D- Palatoglossus

**85- Superior laryngeal nerve supplies motor fibers to.....?**

- A- Epiglottis
- B- All intrinsic laryngeal muscles
- C- Cricothyroid muscle**
- D- Vocal cords

**86- Recurrent laryngeal nerve supplies all intrinsic laryngeal muscles except.....?**

- A- Thyroarytenoid
- B- Cricothyroid**
- C- Lateral cricoarytenoid
- D- Posterior cricoarytenoid

**87- Sensory supply of larynx below vocal cords is by.....?**

- A- Hypoglossal nerve
- B- Glossopharyngeal nerve
- C- Recurrent laryngeal nerve**
- D- Superior laryngeal nerve

**88- Cardiac branches of vagus nerve are....?**

- A- Motor somatic
- B- Parasympathetic**
- C- Sensory
- D- Sympathetic

**89- Auricular branch of vagus supplies skin of.....?**

- A- Mastoid region
- B- Auricle only
- C- External auditory meatus**
- D- Tympanic membrane

**90- Accessory nerve consists of....?**

- A- Facial and spinal roots
- B- Cranial and spinal roots**
- C- Spinal root only
- D- Cranial root only

**91- Cranial root of accessory nerve arises from....?**

- A- Inferior salivatory nucleus
- B- Dorsal motor nucleus
- C- Nucleus ambiguus**
- D- Hypoglossal nucleus

**92- Spinal root of accessory nerve arises from.....?**

- A- Lower 5 cervical segments
- B- Upper 5 cervical segments**
- C- C1–C4
- D- C1–C2

**93- Spinal accessory nerve supplies.....?**

- A- Pharyngeal muscles
- B- Laryngeal muscles
- C- Tongue muscles
- D- Trapezius and sternomastoid**

**94- Injury to spinal accessory nerve in posterior triangle causes.....?**

- A- Loss of taste
- B- Drooping of shoulder**
- C- Tongue deviation
- D- Hoarseness

**95- Cranial part of accessory nerve joins.....?**

- A- Vagus nerve**
- B- Glossopharyngeal nerve
- C- Hypoglossal nerve
- D- Facial nerve

**96- Hypoglossal nerve is.....?**

- A- Parasympathetic
- B- Motor**
- C- Mixed
- D- Sensory

**97- Hypoglossal nerve emerges from the medulla between.....?**

- A- Olive, pons
- B- Pyramid, gracile tubercle
- C- Olive, inferior cerebellar peduncle
- D- Pyramid, olive**

**98- Hypoglossal nerve exits skull through....?**

- A- Stylomastoid foramen
- B- Anterior condylar canal**
- C- Foramen magnum
- D- Jugular foramen

**99- Hypoglossal nerve supplies all tongue muscles except....?**

- A- Palatoglossus**
- B- Styloglossus
- C- Hyoglossus
- D- Genioglossus

**100- Lesion of hypoglossal nerve causes deviation of tongue toward.....?**

- A- No deviation
- B- Paralyzed side**
- C- Opposite side
- D- Normal side

**101- C1 fibers running with hypoglossal nerve supply....?**

- A- Cricothyroid
- B- Stylopharyngeus
- C- Palatoglossus
- D- Thyrohyoid and geniohyoid**

**102- Cervical plexus is formed by ventral rami of.....?**

- A- C3–C6
- B- C2–C5
- C- C1–C4**
- D- C1–C3

**103- Phrenic nerve arises from.....?**

- A- C5–C7
- B- C4–C6
- C- C3–C5**
- D- C2–C4

**104- Cervical plexus is located on the anterior surface of the.....?**

- A- Levator scapulae
- B- Sternomastoid
- C- Scalenus medius**
- D- Scalenus anterior

**105- Descendens cervicalis nerve is formed from.....?**

- A- C3–C5
- B- C2–C3**
- C- C2 only
- D- C1 only

**106- Motor supply to sternomastoid muscle from cervical plexus arises from.....?**

- A- C2–C3**
- B- C3
- C- C2
- D- C11

**107- Which of the following is a sensory branch of the cervical plexus.....?**

- A- Nerve to scalene muscles
- B- Great auricular nerve**
- C- Descendens cervicalis
- D- Phrenic nerve

**108- Supraclavicular nerves supply skin over the.....?**

- A- Shoulder region**
- B- Upper limb
- C- Forehead
- D- Lower limb

## فورماتيف

### وامتحانات سابقة

**109- The extra-ocular muscle which is supplied by the trochlear nerve....?**

A- Superior oblique

B- Inferior rectus

C- Inferior oblique

D- Superior rectus

**110- The nerve supply of the mylohyoid muscle is.....?**

A- Ophthalmic nerve

B- Maxillary nerve

C- Mandibular nerve

D- Facial nerve

**111- How many cranial nerves are classified as strictly sensory nerves....?**

A- 3

B- 4

C- 1

D- 2

**112- Trochlear nerve enters orbit through.....?**

A- Cribriform plate

B- Inferior orbital fissure

C- Superior orbital fissure

D- Optic canal

**113- Types of fibers present in abducent nerve.....?**

A- Special visceral sensory (SVS)

B- Special visceral motor (SVM)

C- General visceral motor (GVM)

D- General somatic motor (GSM)

**114- The seventh cranial nerve supplies....?**

A- Buccinators

B- mylohyoid muscle

C- Masseter

D- Tensor palati

**115- Which of the following cranial nerves is involved in getting the lens of the eye to focus on an object.....?**

A- Sixth cranial nerve

B- Fifth cranial nerve

C- Fourth cranial nerve

D- Third cranial nerve

**116- The skin area over the angle of the mandible is supplied by.....?**

A- Maxillary nerve

B- Mandibular nerve

C- Great auricular nerve

D- Supra-orbital nerve

**117- The nerve supply to the posterior belly of digastrics is.....?**

A- Mandibular nerve

B- Ophthalmic nerve

C- Facial nerve

D- Supra-orbital nerve

**118- Ptosis and lateral squint are features of injury of.....?**

A- Oculo-motor nerve

B- Trochlear nerve

C- Abducent nerve

D- Optic nerve

**119- Chorda tympani nerve joins... fossa?**

A- Buccal nerve

B- Lingual nerve

C- Nerve to masseter

D- Inferior alveolar nerve

**120- The parasympathetic supply of the lacrimal gland is derived from....?**

A- Submandibular ganglion

B- Otic ganglion

C- Pterygo-palatine (Spheno-palatine) ganglion

D- Ciliary ganglion

**121- Ptosis and lateral squint are features of injury of.....?**

- A- Trochlear nerve
- B- Oculo-motor nerve**
- C- Optic nerve
- D- Abducent nerve

**122- Auriculo-temporal nerve is a branch of.....?**

- A- Mandibular nerve**
- B- Facial nerve
- C- Ophthalmic nerve
- D- Maxillary nerve

**123- Infraorbital nerve is a continuation of.....?**

- A- Facial nerve
- B- Mandibular nerve
- C- Maxillary nerve**
- D- Ophthalmic nerve

**124- It leaves the skull through the stylo-mastoid foramen.....?**

- A- Ophthalmic nerve
- B- Facial nerve**
- C- Mandibular nerve
- D- Maxillary nerve

**125- Which of the following cranial nerves can be found in the anterior triangle.....?**

- A- vagus**
- B- Trochlear (CN IV)
- C- Oculomotor (CN III)
- D- Vestibulocochlear

**126- As regards the pathway of tactile sensations from the face, the 1st order neuron relays on the.....?**

- A- Substantia gelatinosa of Rolandi in the posterior horn of spinal cord
- B- Spinal nucleus of trigeminal nerve in the pons, medulla and upper 2 cervical segments
- C- Gracile and cuneate nuclei in the medulla
- D- Main sensory nucleus of trigeminal nerve in the pons**

**127- Sensory nerve of the tympanic cavity is derived from.....?**

- A- Maxillary nerve
- B- Trochlear nerve
- C- Glossopharyngeal nerve**
- D- Facial nerve

**128- Types of fibers present in hypoglossal nerve.....?**

- A- Special visceral sensory (SVS)
- B- General visceral motor (GVM)
- C- Special visceral motor (SVM)
- D- General somatic motor (GSM)**

**129- The Special visceral efferent functional component of the trigeminal nerve is carried by.....?**

- A- Mandibular division**
- B- Ophthalmic division
- C- Maxillary division
- D- Both "ophthalmic division" and "Mandibular division"

**130- Sound and balance are.....?**

- A- Special visceral efferent
- B- Special somatic efferent
- C- Special visceral afferent
- D- Special somatic afferent**

**131- Descend cervicalis arise from.....?**

- A- C4 & C5
- B- C3 & C4
- C- C2 & C3**
- D- C1 & C2

**132- Carotid triangle contain....?**

- A- Mandibular nerve
- B- Facial nerve
- C- Hypoglossal nerve**
- D- Glossopharyngeal nerve

**133- The optic nerve is formed by the convergence of axons from the**

- A- Retinal ganglion cells**
- B- Retinal amacrine cells
- C- Retinal Muller cells
- D- Retinal bipolar cells

**134- Right optic tract contains fibers from.....**

A- The right temporal retina and the left nasal retina.

B- The left temporal retina and the right nasal retina.

C- The right temporal retina only.

D- The left nasal retina only.

**135- Descend cervicalis arise from**

A- C1 & C2

B- C2 & C3

C- C3 & C4

D- C4 & C5

**136- Which of the following branches of the ophthalmic nerve (V1) supplies the nasal cavity.....?**

A- Lacrimal nerve

B- Anterior and posterior ethmoidal nerves

C- Supraorbital nerve

D- Infratrochlear nerve

# Blood supply & lymphatic drainage of the head & neck

**1- Right common carotid artery arises from .....**

- A- common carotid artery
- B- arch of the aorta
- C- brachiocephalic artery**
- D- maxillary

**2- CCA is terminated at the lower border of thyroid cartilage opposite C4**

- A- True
- B- False**

**3- Carotid body and the carotid sinus are associated with .....**

- A- Common carotid artery
- B- External carotid artery
- C- Internal carotid artery
- D- A & B
- E- A & C**

**4- Internal carotid artery is started opposite ..... vertebra**

- A- C2
- B- C3
- C- C4**
- D- C5

**5- Which of the following are branches of anterior branches of ECA?**

- A- superficial temporal & maxillary
- B- Superior thyroid artery & Occipital artery
- C- Facial artery & Lingual artery**
- D- Lingual artery & Ascending pharyngeal artery

**6- Which of the following is ascending and terminal branch of ECA?**

- A- Ascending pharyngeal artery
- B- Occipital artery
- C- Lingual artery
- D- Maxillary artery**

**7- Which of the following maxillary artery branches is incorrectly matched with its supply?**

- A- Deep auricular artery → external auditory canal
- B- Middle meningeal artery → dura mater
- C- Inferior alveolar artery → lower incisors and canines only**
- D- Posterior superior alveolar artery → upper molars and premolars

**8- The third part of the maxillary artery gives rise to all except .....**

- A- Greater palatine artery
- B- Accessory meningeal artery**
- C- Artery to pterygoid canal
- D- Infraorbital artery

**9- Occipital artery is above posterior belly of the digastric muscle**

- A- True
- B- False**

**10- Which of the following part of maxillary artery between the two heads of the lateral pterygoid muscle?**

- A- 1st part
- B- 2nd part
- C- 3rd part**
- D- 4th part

**11- The branch that supplies the hard palate and nasal septum .....**

- A- Greater palatine artery**
- B- Sphenopalatine artery
- C- Posterior superior alveolar artery
- D- Infraorbital artery

**13- Canine & incisors of the upper jaw is supplied by .....**

- A- Infra-orbital artery from 3rd part of subclavian
- B- Posterior superior alveolar artery of maxillary artery
- C- Infra-orbital artery from 2nd part of maxillary artery
- D- Infra-orbital artery from 3rd part of maxillary artery**

**14- Which lymph node group receives lymph from the tip of the tongue?**

- A- Submandibular
- B- Submental**
- C- Retropharyngeal
- D- Jugulo-omohyoid

**15- jugulo-omohyoid lymph nodes is represent .....**

- A- lower deep lateral cervical below superior belly of omohyoid muscle**
- B- upper deep lateral cervical lymph above superior belly of omohyoid muscle
- C- supraclavicular nodes at root of the neck

**16- Which group drains into deep lateral cervical lymph nodes?**

- A- All superficial scalp nodes (occipital, mastoid, pre-auricular)
- B- Buccal nodes only
- C- Submental and submandibular only
- D- All previous groups drain into deep lateral cervical**

**17- Thoracic ducts collect lymph from ..**

- A- right side of upper body
- B- left side of upper body
- C- all the body except right side of upper body**
- D- all the body except left side of upper body

**18- Which of the following arteries supplies the roof of the pharynx?**

- A- Artery to pterygoid canal**
- B- Ascending pharyngeal artery
- C- Greater palatine artery
- D- Deep cervical artery

**19- The lymph node that drains upper lip, cheeks, and anterior tongue is .....**

- A- Submental
- B- Submandibular**
- C- Deep lateral cervical
- D- Retropharyngeal

**20- Superficial lateral cervical lymph nodes drain into**

- A- supraclavicular nodes**
- B- deep lateral cervical lymph nodes

**21- ..... begins as a downward continuation of the sigmoid sinus**

- A- External jugular vein
- B- Subclavian vein
- C- Internal jugular vein**
- D- Anterior jugular vein

**22- The jugular venous arch is located .....**

- A- Between the two internal jugular veins above the hyoid bone
- B- Between the two anterior jugular veins above the jugular notch**
- C- Between the two facial veins in the submental triangle
- D- Between the external and anterior jugular veins behind the sternoclavicular joint

## Formatives

### & Previous Exams

**23- The deep structures of face are supplied by .....**

- A- Superficial temporal artery
- B- Subclavian artery
- C- Maxillary artery**
- D- Fascial artery

**24- EJV terminates by draining into .....**

- A- Retromandibular vein (post branch)
- B- Subclavian vein**
- C- Retromandibular vein (ant branch)
- D- Post auricular vein

**25- Which of the following is a branch from 2nd part of subclavian artery ?**

- A- Internal thoracic
- B- Thyrocervical trunk
- C- Costo-cervical trunk**
- D- axillary artery

**26- Which vein is responsible for draining the majority of the external face?**

- A- Internal jugular vein
- B- Anterior jugular vein
- C- Posterior auricular vein
- D- External jugular vein**

**27- Subclavian vessels are contents of ...**

- A- Muscular triangle
- B- Posterior triangle**
- C- Submental triangle
- D- Digasric triangle

**28- Maxillary artery is divided into three parts by .....**

- A- Temporalis
- B- Masseter
- C- Lateral pterygoid**
- D- Medial pterygoid

**29- Subclavian artery ends at the outer border of the 1st rib by becoming .....**

- A- Axillary**
- B- Internal thoracic
- C- Braciocephalic
- D- Vertebral

**30- The external jugular vein is formed by union of .....**

- A- the anterior division of the retromandibular vein and the posterior auricular vein
- B- the posterior division of the retromandibular vein and the posterior auricular vein**
- C- the anterior division of the retromandibular vein and the occipital vein
- D- the posterior division of the retromandibular vein and the occipital vein

**31- The retro-mandibular vein is formed by union of ...**

- A- Maxillary & superficial temporal veins**
- B- Supra-orbital & supra-trochlear veins
- C- External & internal jugular veins
- D- Internal jugular & subclavian veins

**32- It's a branch of the 3rd part maxillary artery .....**

- A- Infra-orbital artery.**
- B- Middle meningeal artery
- C- Inferior alveolar artery
- D- Masseteric artery

**33- Thyro-cervical trunk arises from .....**

- A- 3rd part subclavian artery
- B- 4th part subclavian artery
- C- 2nd part subclavian artery
- D- 1st part subclavian artery**

**34- The common facial vein ends into .....**

- A- Internal jugular vein**
- B- External jugular vein
- C- Anterior jugular vein
- D- Posterior jugular vein

**35- The deep structures of face are supplied by**

- A- Superficial temporal art
- B- Facial art
- C- Subclavian art
- D- Maxillary art

**36- External jugular vein terminates by draining into**

- A- Post auricular vein
- B- Retromandibular vein (post branche)
- C- Retromandibular vein (ant branche)
- D- Subcalvian vein

# Contents of the orbit

**1- Apex of orbit is located**

- A- Superior
- B- Inferior
- C- Anterior
- D- Posterior

**2- All these bones share in orbit formation except**

- A- Frontal
- B- Nasal
- C- Maxilla
- D- Ethmoid

**3- Which structure of orbit is located at optic foramen**

- A- Apex
- B- Base
- C- Roof
- D- Floor

**4- Lateral wall of the orbit separates the orbit from**

- A- Anterior cranial fossa
- B- Middle cranial fossa
- C- Posterior cranial fossa
- D- Medial cranial fossa

**5- Roof of orbit separates the orbit from**

- A- Anterior cranial fossa
- B- Middle cranial fossa
- C- Posterior cranial fossa
- D- Lateral cranial fossa

**6- The fossa for lacrimal gland is located at of roof**

- A- Anteromedial
- B- Anterolateral
- C- Posterolateral
- D- Postromedial

**7- Which muscle tendon pass through trochlea cartilage**

- A- Superior rectus tendon
- B- Inferior rectus tendon
- C- Superior oblique tendon
- D- Inferior oblique tendon

**8- Floor of orbit separates it from**

- A- Sphenoid air sinus
- B- Ethmoid air sinus
- C- Maxillary air sinus
- D- Frontal air sinus

**9- The strongest wall of the orbit is**

- A- Superior wall
- B- Inferior wall
- C- Medial wall
- D- Lateral wall

**10- All are important landmarks on medial wall of orbit except**

- A- Anterior and posterior ethmoidal foramina
- B- Anterior lacrimal crest
- C- Lacrimal fossa
- D- Lacrimal groove

**11- Anterior lacrimal crest gives attachment to**

- A- Lacrimal sac
- B- Lacrimal part of orbicularis oculi
- C- Medial palpebral ligament
- D- Lateral palpebral ligament

**12- Superior orbital fissure communicates with**

- A- Anterior cranial fossa
- B- Middle cranial fossa
- C- Posterior cranial fossa
- D- Infratemporal fossa

**13- Annulus of zinn gives attachment to**

- A- Infraorbital fossa
- B- Superior orbital fissure**
- C- Inferior orbital fissure
- D- Supraorbital fissure

**14- Superior orbital fissure transmits all of these structures except**

- A- Oculomotor nerve
- B- Superior ophthalmic vein
- C- Nasociliary nerve of ophthalmic
- D- Infraorbital nerve**

**15- Inferior orbital fissure transmits all of these structures except**

- A- Zygomatic nerve
- B- Infraorbital nerve
- C- Orbital branch of middle meningeal artery**
- D- Orbital branches of pterygopalatine ganglion

**16- Common insertion of four recti muscle**

- A- Annulus of zinn
- B- Common tendinous ring
- C- Sclera**
- D- Cornea

**17- Later rectus muscle is supplied by**

- A- Oculomotor
- B- Abducent**
- C- Trochlear
- D- Optic

**18- Superior rectus muscle is supplied by**

- A- Oculomotor**
- B- Optic
- C- Abducent
- D- Trochlear

**19- All these muscle may adduct eye except**

- A- Superior rectus
- B- Inferior rectus
- C- Medial rectus
- D- Lateral rectus**

**20- Which of these muscles is responsible for eye intorsion**

- A- Inferior rectus
- B- Superior rectus
- C- Superior oblique
- D- Inferior oblique
- E- 1 + 3
- F- 2 + 4**

**21- The shortest muscle of the extraocular muscles is**

- A- Superior rectus
- B- Superior oblique
- C- Inferior rectus
- D- Inferior oblique**

**22- Superior oblique muscle is inserted into**

- A- Superior medial quadrant of sclera
- B- Inferior medial quadrant
- C- Inferior lateral quadrant
- D- Superior lateral quadrant**

**23- All are from insertion points of levator palpebrae superioris muscle except**

- A- Superior tarsus
- B- Superior conjunctival fornix
- C- Lesser wing**
- D- Superficial fascia of upper eyelid

**24- Which part of levator palpebrae superioris muscle is responsible for sympathetic tone of eyelid**

- A- Anterior part
- B- Middle part**
- C- Posterior part
- D- Gnabry's part

**25- Sensory innervation of orbit includes all of the following except**

- A- Optic nerve
- B- Zygomatic nerve
- C- Trochlear nerve**
- D- Ophthalmic division of trigeminal

**26- Optic nerve is the cranial nerve number**

- A- I
- B- II**
- C- III
- D- IV

**27- Optic nerve is formed of axons of**

- A- Bipolar cells
- B- Horizontal cells
- C- Ganglion cells**
- D- Amacrine cells

**28- Ophthalmic artery leaves the orbit through**

- A- Superior orbital fissure
- B- Optic foramen**
- C- Inferior orbital fissure
- D- Lacrimal sac

**29- Zygomatic nerve is a branch of**

- A- Optic nerve
- B- Ophthalmic division of trigeminal
- C- Maxillary nerve**
- D- Mandibular nerve

**30- Zygomatic nerve enters orbit through**

- A- Superior orbital fissure
- B- Inferior orbital fissure**
- C- Optic foramen
- D- Lacrimal sac

**31- All are branches of ophthalmic division of trigeminal except**

- A- Frontal nerve
- B- Nasociliary nerve
- C- Inferior orbital nerve**
- D- Lacrimal nerve

**32- Which nerve ends as supraorbital and supratrochlear branches**

- A- Frontal**
- B- Nasociliary
- C- Lacrimal
- D- Inferior orbital

**33- All are branches of nasociliary nerve except**

- A- Long ciliary nerve
- B- Posterior ethmoidal
- C- Anterior ethmoidal
- D- Infraorbital nerve**

**34- Which nerve contains parasympathetic innervation of ciliary and sphincter pupillae muscles**

- A- Oculomotor**
- B- Trochlear
- C- Abducent
- D- Maxillary

**35- Ciliary ganglion is located between optic nerve and**

- A- Medial rectus
- B- Lateral rectus**
- C- Superior oblique
- D- Inferior oblique

**36- All are roots of ciliary ganglion except**

- A- Nasociliary as sensory root
- B- Sympathetic root from plexus around internal carotid artery
- C- Short ciliary nerve form parasympathetic root**
- D- Parasympathetic root from edingerwestphal nucleus

**37- Which one of the following is a branch of ciliary ganglion**

- A- Long ciliary nerve
- B- Short ciliary nerve**
- C- Frontal nerve
- D- Maxillary nerve

**38- Ophthalmic artery arises from**

- A- ECA
- B- ICA**
- C- Subclavin artery
- D- Maxillary artery

**39- Ophthalmic artery gives one these terminal branches**

- A- Supraorbital artery
- B- Supratrochlear artery**
- C- Anterior nasal artery
- D- Lacrimal artery

**40- Ophthalmic artery gives all these branches except**

- A- Anterior ethmoidal artery
- B- Lacrimal artery
- C- Facial artery**
- D- Ciliary artery

**41- Which artery supply pigment epithelium of retina**

- A- Central artery of retina
- B- Ciliary arteries**
- C- Anterior ethmoidal artery
- D- Supraorbital artery

**42- Which ophthalmic vein sends a communicating branch to pterygoid plexus**

- A- Superior ophthalmic
- B- Inferior ophthalmic**
- C- Middle ophthalmic
- D- None of the above

**43- Superior ophthalmic vein connects with**

- A- IJV
- B- EJV
- C- Angular vein**
- D- Retromandibular vein

**44- Eyeball is separated from orbital fat by**

- A- Muller's capsule
- B- Tenon's capsule**
- C- Bowman's capsule
- D- None of the above

**45- Sclera is pierced by optic nerve at**

- A- Lamina ethmoidalis
- B- Lamina cribrosa**
- C- Lamina occipitalis
- D- Lamina frontalis

**46- Which is not true about fibrous coat**

- A- It formed of cornea and sclera
- B- Cornea is avascular structure
- C- Sclera is the largest part of outer coat
- D- Cornea gives attachment to extraocular muscles**

**47- Vascular coat is formed of all the following except**

- A- Choroid
- B- Ciliary body
- C- Cornea**
- D- iris

**48- Which is not true about choroid**

- A- It is highly vascular layer
- B- It supplies inner retina**
- C- It represents the greater part of middle coat
- D- It is pigment layer

**49- Which structure controls shape of lens**

- A- Choroid
- B- Ciliary body**
- C- iris
- D- Cornea

**50- The colored part of the eye is**

- A- Pupil
- B- iris**
- C- Lens
- D- Cornea

**51- Central part of iris contains**

- A- Lens
- B- Pupil**
- C- Retina
- D- Sclera

**52- Which is not true about inner coat**

- A- It is formed of 2 layers
- B- Outer layer is attached to choroid
- C- Pigment layer is deficient in the anterior portion**
- D- Neural layer contains photoreceptors

**53- Macula lutea presents at center of**

- A- Anterior aspect of inner layer
- B- Posterior aspect**
- C- Superior aspect
- D- Inferior aspect

**54- The blind spot area is**

- A- Macula lutea
- B- Fovae centralis
- C- Optic disc**
- D- Macula albicans

**55- Which is not true about lens**

- A- It is transparent avascular structure
- B- Anterior chamber is located between cornea and iris
- C- Posterior segment is filled with aqueous humor**
- D- Ciliary body controls its shape

**56- Which is true about lacrimal apparatus**

- A- Lacrimal gland is formed of 2 parts**
- B- Palpebral part is the larger one
- C- Orbital part is located in lacrimal fossa**
- D- Palpebral part is located at medial part of upper eyelid

**57- Lacrimal apparatus receives parasympathetic supply via**

- A- Glossopharyngeal nerve
- B- Facial nerve**
- C- Trigeminal nerve
- D- Olfactory nerve

**Formatives****& Previous Exams****58- Orbital plate maxillary bone forms**

- A- Roof
- B- Medial wall
- C- Floor**
- D- Lateral wall

**59- The superior orbital fissure is situated between**

- A- Medial wall and roof
- B- Lateral wall and roof**
- C- Medial wall and floor
- D- Lateral wall and floor

**60- The nerve supply of medial rectus muscle is**

- A- Abducent
- B- Oculomotor**
- C- Trochlear
- D- Ophthalmic

**61- Ophthalmic artery enters orbit through**

- A- Optic canal**
- B- Superior orbital fissure
- C- Inferior orbital fissure
- D- Infraorbital foramen

**62- The parasympathetic nerve supply of the lacrimal apparatus comes from**

- A- Oculomotor
- B- Glossopharyngeal
- C- Facial**
- D- Vagus

**63- The branch of ophthalmic artery which supplies the nasal cavity is**

- A- Lacrimal artery
- B- Central retinal artery
- C- Anterior and posterior ethmoidal artery**
- D- Supra orbital

**64- Which muscle is responsible for adduction and lateral rotation of the eyeball**

- A- Superior rectus
- B- Inferior rectus**
- C- Medial rectus
- D- Lateral rectus

**65- The parasympathetic supply of the lacrimal gland is derived from**

- A- Ciliary ganglion
- B- Sphenopalatine ganglion**
- C- Otic ganglion
- D- Submandibular ganglion

# Anatomy of the Ear

**1- Which part of the auricle is composed of soft fibro-fatty tissue?**

- A- Helix
- B- Tragus
- C- Concha
- D- Lobule

**2- The Auriculotemporal nerve supplies which area of the auricle**

- A- The whole cranial surface
- B- The upper part of the lateral surface
- C- The whole lateral surface
- D- The lower part of the lateral surface

**3- The small anterior, superior, and posterior auricular muscles are supplied by which cranial nerve?**

- A- CN V
- B- CN IX
- C- CN VII
- D- CN X

**4- Which part of ear collects and directs sound waves**

- A- Middle ear
- B- External ear
- C- Internal ear
- D- Cochlea

**5- The outer one-third of the external acoustic meatus is:**

- A- Muscular
- B- cartilaginous
- C- Bony
- D- Fibrous

**6- The tympanic membrane separates the external ear from the:**

- A- Inner ear
- B- Eustachian tube
- C- Middle ear
- D- Mastoid air cells

**7- The inner surface of the tympanic membrane is covered by which type of epithelium?**

- A- Stratified squamous epithelium
- B- Simple squamous epithelium
- C- Simple columnar epithelium
- D- Simple cuboidal epithelium

**8- The tympanic membrane lies obliquely with external acoustic meatus at**

- A- 30°
- B- 90°
- C- 55°
- D- 75°

**9- The outer surface of tympanic membrane is ..... while the inner surface is.....**

- A- Concave/convex
- B- Convex/concave

**10- Upper triangular lax part of the tympanic membrane**

- A- Pars tensa
- B- cone of light reflection
- C- Umbo
- D- Pars flaccida

**11- Auricular branch of vagus nerve supply which surface of the tympanic membrane**

- A- Outer surface
- B- Inner surface

**12- The middle ear cavity (tympanic cavity) is located within which bone?**

- A- Sphenoid bone
- B- Occipital bone
- C- Temporal bone
- D- Parietal bone

**13- The anterior wall of the middle ear contains the opening for the:**

- A- Internal acoustic meatus
- B- Vestibule
- C- Eustachian tube**
- D- Semicircular canals

**14- The roof of the middle ear cavity is the:**

- A- Tegmen tympani**
- B- Aditus
- C- Pyramid
- D- Malleus

**15- Round window is closed by**

- A- Stapes
- B- Secondary tympanic membrane**
- C- Malleus
- D- Umbo

**16- Middle ear**

- A- Collect and direct sound waves
- B- Concerned with hearing and equilibrium
- C- Transmit and amplify sound waves**

**17- The floor of the middle ear cavity separates the cavity from the:**

- A- Internal carotid artery
- B- Internal jugular vein**
- C- Facial nerve canal
- D- Mastoid air cells

**18- The three auditory ossicles, in order from medial to lateral, are:**

- A- Incus, Malleus, Stapes
- B- Stapes, Incus, Malleus**
- C- Malleus, Incus, Stapes
- D- Malleus, Stapes, Incus

**19- Covers the oval window**

- A- Malleus
- B- Incus
- C- Stapes**
- D- Secondary tympanic membrane

**20- The muscle that inserted into the handle of the malleus and is supplied by the mandibular nerve is:**

- A- Stapedius
- B- Tensor tympani**
- C- Auricular muscle
- D- Stylopharyngeus

**21- The chorda tympani, a branch of the facial nerve, lies over the inner surface of the:**

- A- Stapes
- B- Promontory
- C- Tympanic membrane**
- D- Oval window

**22- The tympanic plexus, responsible for sensory supply to the middle ear, is formed by branches of which cranial nerve?**

- A- Facial nerve
- B- Trigeminal nerve
- C- Glossopharyngeal nerve**
- D- Vestibulocochlear nerve

**23- Pharyngotympanic tube connects**

- A- External ear with middle ear
- B- Nasopharynx with internal ear
- C- External ear with internal ear
- D- Nasopharynx with middle ear**

**24- The lateral one third of auditory tube is**

- A- Cartilaginous
- B- Fibrous
- C- Bone**
- D- Muscular

**25- The three main components of the bony labyrinth are:**

- A- Auricle, Malleus, Cochlea
- B- Vestibule, Semicircular Canals, Cochlea**
- C- Tympanic membrane, Ossicles, Eustachian tube
- D- Utricle, Saccule, Duct

**26- The membranous labyrinth contains which fluid?**

- A- Perilymph
- B- Endolymph**
- C- Blood
- D- CSF

**27- The snail-shaped structure responsible for hearing is the:**

- A- Vestibule
- B- Semicircular canals
- C- Cochlea**
- D- Utricle

**28- Hollow bony tube in cochlea of inner ear make.....turns around central bony core known as .....**

- A- 2.5/ central petrous
- B- 3.75/ central modiolus
- C- 2.75/ central modiolus**
- D- 1.75/ central umbo

**29- The vestibule houses which two components of the membranous labyrinth?**

- A- Cochlear duct and Sacculle
- B- Utricle and Sacculle**
- C- Utricle and Cochlear duct
- D- Semicircular ducts and Sacculle

**30- Each semicircular duct opens into the utricle via a swelling called the:**

- A- Macula
- B- Ampulla**
- C- Modiolus
- D- Pyramid

**31- The sensory receptors within the ampulla are called the:**

- A- Organ of Corti
- B- Maculae
- C- Cristae ampullaris**
- D- Otoliths

**32- The utricle and saccule contain sensory receptors called the:**

- A- Organ of Corti
- B- Maculae**
- C- Cristae ampullaris
- D- Tectorial membrane

**33- Which structure of the membranous labyrinth is larger and receives the three semicircular ducts?**

- A- Sacculle
- B- Utricle**
- C- Cochlear duct
- D- Ampulla

**34- Communicate the cochlear duct with middle ear cavity**

- A- Oval window
- B- Round window**
- C- Tympanic membrane

**35- The cristae ampullaris are specialized to detect:**

- A- Static head position
- B- Linear acceleration
- C- Angular acceleration**
- D- Sound vibrations

**36- The labyrinthine artery is typically a branch of the:**

- A- External carotid artery
- B- Basilar artery**
- C- Middle meningeal artery
- D- Posterior auricular artery

**37- Origin of stapedius muscle is located in which wall of middle ear**

- A- Medial wall
- B- Floor
- C- Posterior wall**
- D- Anterior wall

**38- What action occurs during swallowing to allow equalization of air pressure on the two sides of the tympanic membrane?**

- A- Contraction of the stapedius muscle
- B- Open auditory tube**
- C- Relaxation of the tensor tympani muscle
- D- Close auditory tube

**39- The medial wall of the vestibule, where it abuts the internal acoustic meatus, is perforated by branches of which nerve?**

- A- Facial nerve
- B- Cochlear nerve
- C- Vestibular nerve**
- D- Glossopharyngeal nerve

**40- Within the cochlear duct, which membrane is described as supporting the spiral organ of Corti?**

- A- Basilar membrane**
- B- Tympanic membrane
- C- Secondary tympanic membrane
- D- Vestibular membrane

## Formatives

### & Previous Exams

**41- The most depressed area of the tympanic membrane corresponds to the tip of the manubrium of handle of malleus.**

- A- Pars tensa
- B- Pars flaccida
- C- Umbo**
- D- Cone of light reflection

**42- A thin plate of bone separating the middle ear cavity from the middle cranial fossa**

- A- Tegmen Tympani**
- B- Mastoid antrum
- C- Pyramid
- D- Processus cochleariformis

**43- The promontory of the 1st coil of cochlea forms....**

- A- Roof
- B- Floor
- C- Medial wall**
- D- Lateral wall

**44- A sensory pain supply of tympanic cavity is mediated by**

- A- Auriculo-temporal nerve 1
- B- Great auricular nerve
- C- Facial nerve
- D- Tympanic plexus of glossopharyngeal nerve**

**49- Which of the following is the middle ear ossicle that is attached to the tympanic membrane?**

- A- Malleus**
- B- Incus
- C- Modiolus
- D- Stapes

**50- Hollow bony tube in cochlea of inner ear makes - turns around central modiolus**

- A- 2 ½**
- B- 1 ½
- C- 3 ½
- D- 2

# Development of head and neck

**1- The flat bones of the skull vault are formed through which process?**

- A- Endochondral ossification
- B- Intramembranous ossification**
- C- Cartilaginous ossification
- D- Chondrocranium formation

**2- Which embryonic structure is responsible for forming the body of the sphenoid bone?**

- A- Ala temporalis
- B- Tubercula cranii
- C- Hypophysial cartilages**
- D- Parachordal plates

**3- The intermaxillary segment is a critical structure that gives rise to:**

- A- The zygomatic bone
- B- The primitive palate and the philtrum**
- C- The definitive palate
- D- The whole lower lip

**4- The crest and tip of the nose are derived from the:**

- A- Frontonasal prominence
- B- Lateral nasal fold
- C- Medial nasal fold**
- D- Maxillary process

**5- A "Posterior cleft palate" is specifically caused by a failure in the fusion of:**

- A- The maxillary process and frontonasal process
- B- The primary palate and secondary palate
- C- The two horizontal palatine processes with the nasal septum**
- D- The two medial nasal folds

**6- Which part of the eye is derived from the surface ectoderm?**

- A- Retina
- B- Iris
- C- Optic nerve
- D- Lens**

**7- The tympanic membrane (eardrum) originates from the:**

- A- First pharyngeal pouch
- B- First pharyngeal membrane**
- C- First pharyngeal cleft
- D- Second pharyngeal arch

**8- Which muscle is the only one derived from the 3rd pharyngeal arch?**

- A- Stapedius
- B- Stylopharyngeus**
- C- Cricothyroid
- D- Temporalis

**9- The recurrent laryngeal nerve (a branch of the vagus nerve) supplies structures derived from which arch?**

- A- 1st arch
- B- 3rd arch
- C- 4th arch
- D- 6th arch**

**10- Parafollicular cells (C-cells) of the thyroid gland originate from the:**

- A- 2nd pharyngeal pouch
- B- 3rd pharyngeal pouch
- C- Ultimobranchial body of the 4th pouch**
- D- Cervical sinus

**11- A branchial fistula is characterized by:**

- A- A closed cyst in the neck
- B- An opening that connects the pharynx with the outside skin**
- C- Failure of the nasolacrimal duct to canalize

**12- "Median cleft lip" is a rare condition caused by:**

- A- Failure of fusion between the maxillary process and medial nasal fold
- B- Failure of development of the medial nasal folds**
- C- Persistence of the cervical sinus
- D- Failure of the palatine shelves to fuse

**13- Pharyngeal apparatus consists of; choose the best statement:**

- A- Pharyngeal arch\_mesoderm / Pharyngeal pouch\_endoderm / Pharyngeal cleft\_ectoderm**
- B- Pharyngeal arch\_ectoderm / Pharyngeal pouch\_endoderm / Pharyngeal cleft\_ectoderm
- C- Pharyngeal arch\_mesoderm / Pharyngeal pouch\_endoderm / Pharyngeal cleft\_endoderm
- D- Pharyngeal arch\_mesoderm / Pharyngeal pouch\_mesoderm / Pharyngeal cleft\_ectoderm

**14- The pharyngeal apparatus is derived from:**

- A- Caudal part of foregut
- B- Cranial part of foregut**
- C- Caudal part of hindgut
- D- Cranial part of hindgut

**15- Meckel's cartilage differentiates into:**

- A- Malleus**
- B- Stapes
- C- Hyoid
- D- Styloid of sphenoid bone

**16- Muscles of mastication are developed from:**

- A- First pharyngeal arch**
- B- Second pharyngeal arch
- C- Reichert's cartilage
- D- First pharyngeal pouch

**17- Mandibular nerve is the nervous element of:**

- A- Mandibular arch
- B- First pharyngeal arch
- C- Meckel's cartilage
- D- All of them are correct**

**18- Middle ear cavity is developed from:**

- A- 1st pharyngeal arch
- B- 1st pharyngeal pouch**
- C- 1st pharyngeal cleft
- D- All of them are correct

**19- Thymus gland is developed from:**

- A- 1st pharyngeal pouch
- B- 2nd pharyngeal pouch
- C- 3rd pharyngeal pouch**
- D- 4th pharyngeal pouch

**20- Forehead is developed from:**

- A- Maxillary process
- B- Frontonasal process**
- C- Mandibular process
- D- All of them are incorrect

**21- Cervical sinus is derived from:**

- A- 2nd pharyngeal cleft
- B- 3rd pharyngeal cleft
- C- 4th pharyngeal cleft
- D- All of them are correct**

**22- Lower lip is developed from:**

- A- Frontonasal process
- B- Maxillary processes
- C- Mandibular processes**
- D- None

**23- Upper parts of cheeks is developed from:**

- A- Frontonasal process
- B- Maxillary processes**
- C- Mandibular processes
- D- None

**24- Greater cornu of hyoid bone derived from:**

- A- First arch
- B- Second arch
- C- Third arch**
- D- Fifth arch

**25- A 43-year-old woman is diagnosed with a condition that causes excruciating pain near her nose and mouth. The involved nerve innervates which branchial arch?**

- A- First**
- B- Second
- C- Third
- D- Fourth

**26- A neonate has a small notch in the transitional zone and vermilion border of the lip. The most likely etiology is:**

- A- Abnormal development of the third and fourth pharyngeal pouches
- B- Bony defects of the molar bone and mandible
- C- Failure of the maxillary processes and medial nasal swellings to fuse**
- D- Incomplete joining of the palatine shelves

**27- Which of the following is a derivative of Reichert cartilage?**

- A- Greater horn of hyoid bone
- B- Malleus
- C- Stapes**
- D- Mandible

**28- Special sense of posterior one third of tongue arises mainly from nerve for:**

- A- First arch
- B- Second arch
- C- Third arch**
- D- Sixth arch

**29- Infant suffering from decreased T-cell function and number is most likely due to defect in:**

- A- 1st pouch
- B- 2nd pouch
- C- 3rd pouch**

**30- Malleus bone is derived from pharyngeal arch; its derivatives are innervated by:**

- A- Trigeminal nerve**
- B- Facial nerve
- C- Glossopharyngeal nerve
- D- Vagus nerve

## Formatives

**31- Which cranial nerve provides motor innervation to muscles of mastication?**

- A- Ophthalmic division of trigeminal (V1)
- B- Maxillary division of trigeminal (V2)
- C- Mandibular division of trigeminal (V3)**
- D- Facial nerve (VII)

**32- The superior parathyroid gland is derived from:**

- A- 1st pharyngeal arch
- B- 3rd pharyngeal pouch
- C- 4th pharyngeal pouch**
- D- 5th pharyngeal pouch

**33- Muscles of mastication are derivatives of:**

- A- 1st pharyngeal arch**
- B- 2nd pharyngeal arch
- C- 3rd pharyngeal arch
- D- 4th pharyngeal arch

**34- Facial nerve is the nerve component of:**

- A- 1st pharyngeal arch
- B- 2nd pharyngeal arch**
- C- 3rd pharyngeal arch
- D- 4th pharyngeal arch

# Brainstem

**1- The widest and most superior part of the brainstem is**

- A- Medulla oblongata
- B- Pons
- C- Midbrain**
- D- Cerebellum

**2- The distal part of the brain connects diencephalon with spinal cord**

- A- Cerebrum
- B- Cerebellum
- C- Brain stem**
- D- Spinal cord

**3- The narrowest and most caudal part of the brainstem is**

- A- Midbrain
- B- Pons
- C- Medulla oblongata**
- D- Cervical spinal cord

**4- The brainstem extends from:**

- A- Optic chiasma to foramen magnum
- B- Tentorial aperture to upper border of pons
- C- just above Aperture in tentorium cerebelli to just below foramen magnum**
- D- From cerebral cortex to LGB

**5- The cerebellum is related to the brainstem mainly on its**

- A- Ventral surface
- B- Lateral surface
- C- Dorsal surface**
- D- Inferior surface

**6- The medulla oblongata extends from**

- A- Inferior pontine sulcus superiorly to decussation of pyramids inferiorly**
- B- Superior pontine sulcus to spinal cord
- C- Foramen magnum to superior colliculus
- D- Inferior colliculus to pons

**7- The medullary striae separate**

- A- Medulla from spinal cord
- B- Back of medulla from back of pons**
- C- Ventral surface from dorsal surface of medulla
- D- Closed from open medulla

**8- Which part of the medulla contains the central canal?**

- A- Closed medulla**
- B- Upper medulla
- C- Open medulla
- D- Tegmentum

**9- Open medulla is related posteriorly to**

- A- Lt Lateral ventricle
- B- 3rd ventricle
- C- 4th ventricle**
- D- Rt lateral ventricle

**10- Medulla is divided into 2 halves anteriorly by**

- A- Anterior medial fissure
- B- Anterior lateral fissure
- C- Anterior median fissure**
- D- Dorsal fissure

**11- How many elevations lie on one side of anterior median fissure**

- A- 3**
- B- 4
- C- 6
- D- 8

**12- The nearest elevation to midline of medulla**

- A- Pyramid**
- B- Olive
- C- Inferior cerebellar peduncle
- D- Superior cerebellar peduncle

**13- The olive is formed by**

- A- Inferior cerebellar peduncle
- B- Superior olivary nucleus
- C- Inferior olivary nucleus**
- D- Vestibular nuclei

**14- The inferior cerebellar peduncle connects the**

- A- Pons to cerebellum
- B- Midbrain to cerebellum
- C- Medulla to cerebellum**
- D- Spinal cord to cerebellum

**15- The most lateral elevations of medulla anteriorly**

- A- Pyramid
- B- Olive
- C- Inferior cerebellar peduncle**
- D- Superior cerebellar peduncle

**16- Ventral aspect of medulla gives exit of**

- A- Upper 4 cranial nerves
- B- Middle 4 cranial nerves
- C- Lower 4 cranial nerves**
- D- All cranial nerves

**17- The hypoglossal nerve exits the medulla through**

- A- Posterolateral sulcus between olive and inferior cerebellar peduncle
- B- Anterolateral sulcus between pyramid and olive**
- C- Between Anterior median fissure and pyramid
- D- Between 2 Inferior cerebellar peduncle

**18- Which cranial nerves emerge from the posterolateral sulcus of the medulla?**

- A- IX, X, XI**
- B- VII, VIII
- C- V, VI
- D- III, IV

**19- All these elevations lie on dorsal aspect of closed medulla except**

- A- Gracile tubercle
- B- Cuneate tubercle
- C- Vestibular trigone**
- D- Tuberculum cinereum

**20- The cuneate tubercle lies**

- A- Medial to gracile tubercle
- B- Lateral to gracile tubercle**
- C- Lateral to tuberculum cinereum
- D- In open medulla

**21- Tuberculum cinereum is formed by**

- A- Cuneate nucleus
- B- Vestibular nuclei
- C- Spinal nucleus of trigeminal nerve**
- D- Inferior olivary nucleus

**22- Open medulla forms lower part of floor of**

- A- Lateral ventricle
- B- 3rd ventricle
- C- 4th ventricle**
- D- Medial ventricle

**23- Which structure separates the 3 trigones of open medulla**

- A- Superior fovea
- B- Inferior fovea**
- C- Medial fovea
- D- Lateral fovea

**24- All these nerves share in dorsal aspect of open medulla except**

- A- XII
- B- X
- C- XI**
- D- VIII

**25- Which trigone is the most lateral in the floor of the 4th ventricle**

- A- Hypoglossal
- B- Vagal
- C- Vestibular**
- D- Facial

**26- About 85% of corticospinal fibers decussate at**

- A- Upper pons
- B- Lower pons
- C- Lower closed medulla**
- D- Upper closed medulla

**27- The sensory decussation is formed by**

- A- Corticospinal fibers
- B- External arcuate fibers
- C- Internal arcuate fibers**
- D- Spinothalamic fibers

**28- Sensory decussation is related to pyramidal decussation**

- A- Superior**
- B- Inferior
- C- Medial
- D- Lateral

**29- Fine touch and proprioception from the lower half of the body are conveyed by**

- A- Cuneate nucleus
- B- Gracile nucleus**
- C- Spinal nucleus of trigeminal
- D- Inferior olivary nucleus

**30- Inferior olivary nucleus is located on open medulla**

- A- Superior to pyramid
- B- Inferior to pyramid
- C- Lateral to pyramid**
- D- Medial to pyramid

**31- The inferior olivary nucleus is mainly concerned with**

- A- Olfactory processing
- B- Auditory processing
- C- Motor coordination & learning**
- D- Visual processing

**32- Fibers from inferior olivary nucleus enter cerebellum through**

- A- Superior cerebellar peduncle
- B- Middle cerebellar peduncle
- C- Inferior cerebellar peduncle**
- D- Crus cerebri

**33- Hypoglossal nucleus gives supply to all these muscle except**

- A- Hypoglossus
- B- Hyoglossus
- C- Palatoglossus**
- D- Genioglossus

**34- Motor fibers to pharynx and larynx arise from**

- A- Hypoglossal nucleus
- B- Inferior salivatory nucleus
- C- Nucleus ambiguus**
- D- Solitary nucleus

**35- Nucleus ambiguus gives motor fibers to all these cranial nerves except**

- A- 9th
- B- 10th
- C- 11th
- D- 12th**

**36- Nucleus ambiguus gives supply to all palate muscle except**

- A- Tensor palate**
- B- Levator palati
- C- Palatoglossus
- D- Palatopharyngeus

**37- Taste sensation from tongue is received by**

- A- Inferior salivatory nucleus
- B- Spinal nucleus of trigeminal
- C- Nucleus solitarius**
- D- Vestibular nuclei

**38- Which one of these lies lateral to hypoglossal nucleus**

- A- Nucleus ambiguus
- B- Dorsal motor nucleus of vagus**
- C- Spinal nucleus of trigeminal
- D- Nucleus solitarius

**39- Carries pain and temperature sensation from the face**

- A- Hypoglossal nucleus
- B- Nucleus ambiguus
- C- Spinal nucleus of trigeminal**
- D- Dorsal motor nucleus of vagus

**40- Nucleus solitarius receives taste fibers from all these nerves except**

- A- VII
- B- IX
- C- X
- D- XI**

**41- Inferior salivatory nucleus supplies parotid gland via**

- A- Facial
- B- Hypoglossal
- C- Glossopharyngeal**
- D- Vagus

**42- Which of these nuclei is parasympathetic**

- A- Hypoglossal nucleus
- B- Nucleus ambiguus
- C- Dorsal motor nucleus of vagus**
- D- Spinal nucleus of trigeminal

**43- The pons is located**

- A- Between midbrain & spinal cord
- B- Between midbrain & medulla**
- C- Between cerebellum & thalamus
- D- Anterior to spinal cord

**44- The basilar artery lies in**

- A- Pontine sulcus
- B- Basilar groove**
- C- Median fissure
- D- Cerebellopontine angle

**45- Transverse pontine fibers form**

- A- Inferior cerebellar peduncle
- B- Superior cerebellar peduncle
- C- Middle cerebellar peduncle**
- D- Crus cerebri

**46- The trigeminal nerve emerges from**

- A- Midline of pons
- B- Posterior surface
- C- Anterolateral surface of pons**
- D- Cerebellopontine angle

**47- Which structure is NOT a content of cerebellopontine angle?**

- A- Facial nerve
- B- Vestibulocochlear nerve
- C- Anterior inferior cerebellar artery
- D- Oculomotor nerve**

**48- Which nerve emerges between pons and pyramid**

- A- Trigeminal
- B- Abducent**
- C- Facial
- D- Vestibulocochlear

**49- Upper half of the floor of the 4th ventricle is formed of**

- A- Upper medulla
- B- Lower medulla
- C- Ventral aspect of pons
- D- Dorsal aspect of pons**

**50- Pons is related lateral to**

- A- Superior cerebellar peduncle**
- B- Middle cerebellar peduncle
- C- Inferior cerebellar peduncle
- D- Crus cerebri

**51- Medial eminence on dorsal aspect of pons is produced by**

- A- Facial
- B- Abducent**
- C- Glossopharyngeal
- D- Vagus

**52- Middle 4 cranial nerves exit from**

- A- Medulla
- B- Pons
- C- Midbrain
- D- Cerebellum

**53- Facial colliculus is formed by**

- A- Facial nucleus
- B- Abducent nucleus alone
- C- Facial nerve looping around abducent nucleus
- D- Vestibular nuclei

**54- The anterior part of pons is called**

- A- Tegmentum
- B- Basis pontis
- C- Tectum
- D- Crus cerebri

**55- Transverse pontocerebellar fibers cross to opposite side to**

- A- Superior cerebellar peduncle
- B- Middle cerebellar peduncle
- C- Inferior cerebellar peduncle
- D- Crus cerebri

**56- Tegmentum of pons contains all of the following except**

- A- Nuclei of middle 4 cranial nerves
- B- Ascending lemniscus
- C- Pontine reticular formation
- D- Pontine cerebellar fibers

**57- Motor nucleus of trigeminal nerve is located in**

- A- Medulla
- B- Midbrain
- C- Lower pons
- D- Mid-level of pons

**58- Motor nucleus of trigeminal nerve supplies all EXCEPT**

- A- Muscles of mastication
- B- Mylohyoid
- C- Anterior belly of digastric
- D- Posterior belly of digastric

**59- Main sensory nucleus of trigeminal nerve**

- A- Lies medial to motor nucleus
- B- Lies lateral to motor nucleus
- C- Lies in medulla
- D- Lies in midbrain

**60- Spinal nucleus of trigeminal**

- A- Lies only in pons
- B- Extends downward into medulla
- C- Lies in midbrain
- D- Lies lateral to facial nucleus

**61- It receives touch sensation from the same side of face**

- A- Motor nucleus of trigeminal
- B- Main sensory nucleus of trigeminal
- C- Spinal nucleus of trigeminal
- D- Mesencephalic nucleus

**62- Spinal nucleus of trigeminal receives**

- A- Touch
- B- Proprioception
- C- Pain and temperature
- D- Taste

**63- Which trigeminal nucleus contains first-order sensory neurons?**

- A- Main sensory nucleus
- B- Spinal nucleus
- C- Mesencephalic nucleus
- D- Motor nucleus

**64- Which nucleus of trigeminal extends into 2 different parts of brainstem**

- A- Motor nucleus
- B- Spinal nucleus
- C- Mesencephalic nucleus
- D- Main sensory nucleus

**65- Motor nucleus of abducent nerve lies**

- A- In lower pons
- B- In midbrain
- C- In medulla
- D- In upper pons

**66- Abducent nerve supplies**

- A- Medial rectus
- B- Inferior rectus
- C- Lateral rectus**
- D- Superior rectus

**67- Motor nucleus of facial nerve is located in**

- A- Upper pons
- B- Lower pons**
- C- Medulla
- D- Midbrain

**68- Facial nerve fibers loop around which nucleus to form facial colliculus**

- A- Oculomotor nucleus
- B- Trigeminal nucleus
- C- Abducent nucleus**
- D- Vestibular nucleus

**69- Upper half of facial nucleus receives**

- A- Contralateral corticobulbar fibers
- B- Ipsilateral corticobulbar fibers
- C- Bilateral corticobulbar fibers**
- D- No corticobulbar fibers

**70- Nucleus solitarius lies in**

- A- Pons
- B- Medulla**
- C- Midbrain
- D- Cerebellum

**71- Taste fibers from anterior 2/3 of tongue reach nucleus solitarius via**

- A- Glossopharyngeal nerve
- B- Vagus nerve
- C- Chorda tympani**
- D- Hypoglossal nerve

**72- Superior salivatory nucleus is located in**

- A- Medulla
- B- Lower pons**
- C- Upper pons
- D- Midbrain

**73- Superior salivatory nucleus supplies**

- A- Parotid gland
- B- Sublingual gland
- C- Submandibular gland
- D- B & C**

**74- Number of vestibular nuclei is**

- A- Two
- B- Three
- C- Four**
- D- Five

**75- Superior and lateral vestibular nuclei are located in**

- A- Medulla
- B- Lower pons**
- C- Upper pons
- D- Midbrain

**76- Inferior and medial vestibular nuclei are located in**

- A- Pons
- B- Midbrain
- C- Medulla**
- D- Cerebellum

**77- It is 1st neuron in central auditory system for binaural hearing**

- A- Inferior olivary nucleus
- B- Superior olivary nucleus**
- C- Superior salivatory nucleus
- D- Inferior salivatory nucleus

**78- Ventral and dorsal cochlear nuclei are located in**

- A- Medulla
- B- Lower pons**
- C- Upper pons
- D- Midbrain

**79- Which cochlear nucleus forms trapezoid body by its axon mainly**

- A- Ventral cochlear nucleus**
- B- Dorsal cochlear nucleus

**80- The most lateral lemniscus is**

- A- Medial lemniscus
- B- Trigeminal lemniscus
- C- Spinal lemniscus
- D- Lateral lemniscus**

**81- Lateral lemniscus carries**

- A- Gustatory pathway
- B- Auditory pathway**
- C- Motor pathway
- D- Visual pathway

**82- Lateral lemniscus carries hearing from**

- A- One ear only
- B- Both ears**
- C- Same side ear
- D- Opposite ear

**83- Lateral lemniscus originates from**

- A- Gracile nucleus
- B- Cochlear nuclei & superior olivary nucleus**
- C- Vestibular nuclei
- D- Inferior colliculus

**84- Lateral lemniscus ends in**

- A- Superior colliculus
- B- Inferior colliculus & MGB**
- C- Thalamus only
- D- Cerebellum

**85- Spinal lemniscus carries**

- A- Fine touch
- B- Proprioception
- C- Pain, temperature & crude touch**
- D- Hearing

**86- Spinal lemniscus ends in**

- A- PMVNT
- B- PLVNT**
- C- Medial nucleus
- D- Ventral nucleus

**87- Spinal lemniscus originates from axons of**

- A- Spinal nucleus and main sensory nucleus
- B- SGR and nucleus proprius**
- C- Gracile and cuneate nuclei
- D- Cochlear nuclei

**88- Trigeminal lemniscus carries sensation from**

- A- Same side of body
- B- Same side of face
- C- Opposite side of face**
- D- Opposite side of body

**89- Trigeminal lemniscus ends in**

- A- PLVNT
- B- PMVNT**
- C- LGN
- D- MGB

**90- Medial lemniscus originates from**

- A- Trigeminal nuclei
- B- Cochlear nuclei
- C- Gracile & cuneate nuclei**
- D- SGR and nucleus proprius

**91- Medial lemniscus carries**

- A- Pain and temperature
- B- Hearing
- C- Proprioception and fine touch**
- D- Vision

**92- Medial lemniscus ends in**

- A- PMVNT
- B- PLVNT**
- C- LGN
- D- MGB

**93- The midbrain passes through opening in**

- A- Falx cerebri
- B- Tentorium cerebelli**
- C- Falx cerebelli
- D- Optic canal

**94- Midbrain is connected to cerebellum by**

- A- Inferior cerebellar peduncles
- B- Middle cerebellar peduncles
- C- Superior cerebellar peduncles**
- D- All of the above

**95- Each half of the midbrain forms**

- A- Crus cerebelli
- B- Cerebral peduncle**
- C- Pyramid
- D- Olive

**96- Emerges from interpeduncular fossa and lies medial to crus cerebri**

- A- Trochlear
- B- Oculomotor**
- C- Abducent
- D- Glossopharyngeal

**97- Superior colliculi are centers for**

- A- Auditory reflexes
- B- Visual reflexes**
- C- Pain sensation
- D- Motor coordination

**98- Inferior colliculi are concerned with**

- A- Vision
- B- Balance
- C- Hearing**
- D- Proprioception

**99- Trochlear nerve emerges**

- A- From interpeduncular fossa
- B- Below inferior colliculus dorsally**
- C- From pons
- D- From medulla

**100- Superior brachium connects**

- A- Superior colliculus to lateral geniculate body**
- B- Inferior colliculus to medial geniculate body
- C- Superior colliculus to medial geniculate body
- D- Inferior colliculus to lateral geniculate body

**101- Which of these structures don't wind around lateral side of midbrain**

- A- Posterior cerebral arteries
- B- Superior cerebellar arteries
- C- Oculomotor nerve**
- D- Optic tract

**102- Anterior part of cerebral peduncle is**

- A- Tegmentum
- B- Substantia nigra
- C- Crus cerebri**
- D- Tectum

**103- Which of these fibers pass through middle 3/5 of cerebral peduncle**

- A- Medial corticobulbar tract
- B- Corticospinal tract**
- C- Temporo-pontine
- D- Lateral corticobulbar tract
- E- Fronto-pontine

**104- Substantia nigra separates**

- A- Crus cerebri from tectum
- B- Crus cerebri from tegmentum**
- C- Tegmentum from tectum
- D- Cerebral aqueduct from tectum

**105- Neurotransmitter secreted by pars compacta is**

- A- GABA
- B- Acetylcholine
- C- Dopamine**
- D- Serotonin

**106- Degeneration of substantia nigra leads to**

- A- Chorea
- B- Parkinsonism**
- C- Huntington disease
- D- Hemiballismus

**107- Motor nucleus of oculomotor lies at level of**

- A- Superior colliculus**
- B- Inferior colliculus

**108- Edinger–Westphal nucleus supplies**

- A- Dilator pupillae
- B- Sphincter pupillae**
- C- Extraocular muscles
- D- Lacrimal gland

**109- Trochlear nucleus is located at the level of**

- A- Superior colliculus
- B- Inferior colliculus**

**110- Red nucleus is located in**

- A- Crus cerebri
- B- Tegmentum**
- C- Tectum
- D- Substantia nigra

**111- Red nucleus is an**

- A- Pyramidal center
- B- Sensory nucleus
- C- Extraparamidal center**
- D- Autonomic center

**112- Red nucleus receives input from all EXCEPT**

- A- Cerebral cortex
- B- Cerebellum
- C- Basal ganglia
- D- Spinal cord**

**113- Main output of red nucleus is via**

- A- Corticospinal tract
- B- Tectospinal tract
- C- Rubrospinal tract**
- D- Vestibulospinal tract

**114- Controls crawling in babies**

- A- Substantia nigra
- B- Red nucleus**

**115- Superior colliculi are connected together by**

- A- Anterior commissure
- B- Corpus callosum
- C- Posterior commissure**
- D- Habenular commissure

**116- Afferent fibers to superior colliculus include**

- A- Optic nerve
- B- Visual cortex
- C- Spinotectal tract
- D- All of the above**

**117- Control of conjugate eye movement is a function of**

- A- Inferior colliculus
- B- Superior colliculus**
- C- Red nucleus
- D- Substantia nigra

**118- The center of Pupillary light reflex is**

- A- Lateral geniculate body
- B- Superior colliculus
- C- Pretectal nuclei**
- D- Edinger-Westphal nucleus

**119- Anterior boundary of interpeduncular fossa is**

- A- Optic tracts
- B- Optic chiasma**
- C- Cerebral peduncles
- D- Upper border of pons

**120- Anterolateral boundaries of interpeduncular fossa are**

- A- Optic nerves
- B- Optic tracts**
- C- Cerebral peduncles
- D- Mammillary bodies

**121- All are contents of interpeduncular fossa EXCEPT**

- A- Oculomotor nerve
- B- Circle of Willis
- C- Interpeduncular cistern
- D- Trochlear nerve**

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- D- Hemiballismus

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- A- Superior colliculus**
- B- Inferior colliculus

## Formatives

### & Previous Exams

**137- Fibers from nucleus gracilis and cuneatus decussate and form:**

- A- Medial lemniscus
- B- Lateral lemniscus
- C- Trigeminal lemniscus
- D- Spinal lemniscus

**138- Tumor in cerebellopontine angle affects:**

- A- Oculomotor nerve
- B- Trigeminal nerve
- C- Facial nerve
- D- Abducent nerve

**139- The most medial fibers of crus cerebri are:**

- A- Corticospinal fibers
- B- Frontopontine fibers
- C- Temporopontine fibers
- D- Corticobulbar fibers

**140- Cerebellum is connected with midbrain by:**

- A- Cerebral peduncles
- B- Corpus callosum
- C- Fornix
- D- Superior cerebellar peduncle

**141- These nerves are seen in the ponto-cerebellar angle:**

- A- Glossopharyngeal & vagus nerves
- B- Vagus & Accessory nerves
- C- Facial & vestibulocochlear nerves

**142- A diamond-shaped space in the base of forebrain, bounded by optic chiasm and optic tract anterior & cerebral peduncles posterior:**

- A- Fourth ventricle
- B- Interpeduncular fossa
- C- Rhomboid fossa
- D- Cavernous sinus

**143- Blackish coloration, due to its high content of polymerized dopamine:**

- A- Globus Pallidus
- B- Putamen
- C- Substantia Nigra
- D- Red nucleus

**144- The 1st station of binaural hearing is:**

- A- Spiral ganglion
- B- Cochlear nuclei
- C- Inferior colliculus
- D- Superior olivary nucleus

**145- Inferior quadrigeminal brachium conveys fibers from:**

- A- Inferior colliculus to medial geniculate body
- B- Inferior colliculus to lateral geniculate body
- C- Superior colliculus to medial geniculate body
- D- Superior colliculus to lateral geniculate body

**146- The facial colliculus is a bulging formed by:**

- A- Facial fibers and facial nucleus
- B- Facial fibers and abducent nucleus
- C- Abducent fibers and facial nucleus
- D- Abducent fibers and abducent nucleus

**147- Cranial nerve 4 (Trochlear) is located inferior to what landmark:**

- A- Superior colliculus
- B- Inferior colliculus
- C- Lateral colliculus
- D- Medial colliculus

**148- The posterior white matter of medulla oblongata medially contains:**

- A- Fasciculus gracilis
- B- Fasciculus cuneatus
- C- Nucleus gracilis
- D- Nucleus cuneatus

**149- Cerebro-ponto-cerebellar tract is connected to cerebellum by:**

- A- Superior cerebellar peduncle
- B- Middle cerebellar peduncle
- C- Inferior cerebellar peduncle
- D- None of the above

# Histological structure of Neuron, Synapse, Neuroglial Cells

## 1- Nervous tissue consists mainly of.....?

- A- Muscle fibers and connective tissue
- B- Connective tissue membranes
- C- Epithelium and glands
- D- Neurons and neuroglia**

## 2- The central nervous system includes....?

- A- Nerve axons and ganglia
- B- Brain and spinal cord**
- C- Epithelium and glands
- D- Connective tissue membranes

## 3- The structural and functional unit of the nervous system is the.....?

- A- Neuroglial cell
- B- Synapse
- C- Neuron**
- D- Schwann cell

## 4- Neurons are considered permanent cells because they are in.....?

- A- M phase
- B- G0 phase**
- C- S phase
- D- G1 phase

## 5- Neuron is composed of.....?

- A- Cell body and nucleus only
- B- Cell body and processes**
- C- Dendrites only
- D- Axon only

## 6- The cell body of a neuron is also called the.....?

- A- Endoneurium
- B- Neurolemma
- C- Perikaryon**
- D- Axoplasm

## 7- Under light microscope, the neuron nucleus is typically.....?

- A- Large, central, pale with prominent nucleolus**
- B- Peripheral with condensed chromatin
- C- Small and dark
- D- Absent

## 8- Nissl granules are mainly composed of.....?

- A- Mitochondria
- B- Rough endoplasmic reticulum**
- C- Smooth endoplasmic reticulum
- D- Golgi apparatus

## 9- Nissl granules are present in all of the following except.....?

- A- Perikaryon
- B- Axon**
- C- Dendrites
- D- Cell body

## 10- Lipofuscin pigment represents.....?

- A- Neurotransmitter storage
- B- Melanin synthesis
- C- Wear and tear pigment**
- D- Secretory product

## 11- Melanin pigment is characteristically found in.....?

- A- Retina
- B- Spinal cord
- C- Substantia nigra**
- D- Cerebellum

## 12- Under electron microscope, neuronal chromatin is mainly...?

- A- Absent
- B- Euchromatic**
- C- Condensed
- D- Heterochromatic

**13- The cytoskeleton of neurons consists mainly of.....?**

- A- Collagen fibers
- B- Neurofilaments and neurotubules**
- C- Microfilaments only
- D- Actin only

**14- The axon is best described as...?**

- A- Short and broad
- B- Multiple and unbranched
- C- Single, long, and thin**
- D- Multiple, short, and thick

**15- The main function of the axon is to.....?**

- A- Form synapses only
- B- Produce CSF
- C- Transmit nerve impulses**
- D- Receive impulses

**16- Axoplasm contains all of the following except.....?**

- A- Mitochondria
- B- RER**
- C- Neurofilaments
- D- Microtubules

**17- Dendrites are characterized by being.....?**

- A- Myelinated
- B- Non-branched
- C- Multiple, short, and branched**
- D- Single and long

**18- Which structure receives nerve impulses.....?**

- A- Synaptic cleft
- B- Node of Ranvier
- C- Dendrite**
- D- Axon

**19- The most common type of neuron is.....?**

- A- Anaxonic
- B- Multipolar**
- C- Pseudounipolar
- D- Bipolar

**20- Anterior horn cells are examples of.....?**

- A- Unipolar neurons
- B- Multipolar neurons**
- C- Pseudounipolar neurons
- D- Bipolar neurons

**21- Golgi type I neurons are characterized.....?**

- A- No dendrites
- B- No axons
- C- Long axons**
- D- Short axons

**22- Bipolar neurons are commonly found in.....?**

- A- Cerebral cortex
- B- Retina**
- C- Autonomic ganglia
- D- Spinal ganglia

**23- Pseudounipolar neurons are mainly found in....?**

- A- Brain stem
- B- Spinal ganglia**
- C- Cerebellum
- D- Autonomic ganglia

**24- Interneurons mainly function to....?**

- A- Control muscles
- B- Form myelin
- C- Modify responses of other neurons**
- D- Conduct sensory impulses

**25- Neuroglia in the CNS replace connective tissue and function to...?**

- A- Form synapses only
- B- Produce hormones
- C- Support neurons**
- D- Conduct impulses

**26- Myelination in the CNS is done by....?**

- A- Microglia
- B- Oligodendrocytes**
- C- Astrocytes
- D- Schwann cells

**27- Microglia function mainly as....?**

- A- CSF producing cells
- B- Macrophages**
- C- Supporting cells
- D- Insulating cells

**28- Astrocytes are involved in....?**

- A- CSF circulation only
- B- Phagocytosis only
- C- Formation of blood brain barrier**
- D- Myelination in PNS

**29- Ependymal cells are responsible for....?**

- A- Synaptic transmission
- B- Lining ventricles and CSF formation**
- C- Phagocytosis
- D- Myelin formation

**31- A synapse is best defined as.....?**

- A- Supporting cell
- B- Specialized junction for signal transmission**
- C- Nerve fiber
- D- Neuron cell body

**32- The presynaptic membrane is formed by....?**

- A- Synaptic cleft
- B- Terminal button of axon**
- C- Cell body
- D- Dendrite

**33- The most common type of synapse is.....?**

- A- Mechanical
- B- Chemical**
- C- Mixed
- D- Electrical

**34- Electrical synapses are characterized by.....?**

- A- Absence of ions
- B- Gap junctions**
- C- Slow transmission
- D- Neurotransmitter release

**35- Which feature is characteristic of autonomic ganglia.....?**

- A- Grouped cell bodies
- B- Numerous synapses**
- C- Numerous satellite cells
- D- Large pseudounipolar neurons

**36- Which feature is characteristic of Spinal ganglia....?**

- A- Large pseudounipolar neurons
- B- Numerous satellite cells
- C- Grouped cell bodies
- D- All of above**

**Schwann cells are found in the.....?**

- A- Brain ventricles
- B- PNS only**
- C- CNS only
- D- Cerebral gray matter

## فورماتيف

### وامتحانات سابقة

**37- The site of contact between two neurons is called.....?**

- A- Dendrites
- B- Cell body
- C- Synapse**
- D- Axons

**38- When drug molecules pass from the blood to the neuron which of the following cells will control their passage.....?**

- A- Microglia
- B- Schwann cells
- C- Astrocytes**
- D- Oligodendrocyte

**39- Which of the following cells are responsible for Myelination of CNS nerve fibers...?**

- A- Ependymal cells
- B- Oligodendrocytes**
- C- Schwann cells
- D- Microglia

**40- Which of the followings are the phagocytic cells of the brain.....?**

- A- Ependymal cells
- B- Oligodendrocytes
- C- Schwann cells
- D- Microglia**

**41- What is meant by the point of Connection between one neuron to another.....?**

- A- Nucleus
- B- Synapse**
- C- Neuroglia
- D- Ganglia

**42- Which of the following statements best describes neurons of sympathetic ganglia.....?**

- A- Their nerve fibers are mostly myelinated
- B- They are pseudounipolar
- C- They are supported by microglia
- D- They are small and scattered**

**43- Which of the following is found in the peripheral nervous system?**

- A- Astrocytes
- B- Schwann cell**
- C- Oligodendrocytes
- D- Microglia

# Central Nervous system

**1- The meninges are best defined as.....?**

- A- Neuroglial cells
- B- Epithelial layers lining ventricles
- C- Connective tissue membranes protecting the CNS**
- D- Nervous tissue layers

**2- The meninges protect the.....?**

- A- Peripheral nerves
- B- Brain and spinal cord**
- C- Spinal cord only
- D- Brain only

**3- The outermost meningeal layer is the.....?**

- A- Ependyma
- B- Dura mater**
- C- Arachnoid mater
- D- Pia mater

**4- Dura mater is mainly composed of....?**

- A- Simple squamous epithelium
- B- Dense fibro-elastic connective tissue**
- C- Nervous tissue
- D- Loose connective tissue

**5- The dura mater is strongly adherent to the....?**

- A- Periosteum of the skull**
- B- Arachnoid mater
- C- Brain surface
- D- Pia mater

**6- The arachnoid mater is described as...?**

- A- Stratified epithelium
- B- Muscular
- C- Cobweb-like**
- D- Highly vascular

**7- The layer of arachnoid mater in contact with dura is composed of...?**

- A- Stratified squamous epithelium
- B- Simple squamous epithelium**
- C- Simple cuboidal epithelium
- D- Web like strands of CT like pillars

**8- The subarachnoid space contains all except....?**

- A- Lymphatic vessels**
- B- Veins
- C- Arteries
- D- CSF

**9- All surfaces of the arachnoid mater are covered by....?**

- A- Neuroglial cells
- B- Simple squamous epithelium**
- C- Endothelium
- D- Cuboidal epithelium

**10- The innermost meningeal layer is the....?**

- A- Ependyma
- B- Pia mater**
- C- Arachnoid mater
- D- Dura mater

**11- Pia mater is best described as...?**

- A- Nervous tissue
- B- Stratified epithelium
- C- Loose vascular CT**
- D- Dense avascular CT

**12- Pia mater adheres closely to the surface of the.....?**

- A- Peripheral nerves
- B- Ventricles only
- C- Brain and spinal cord**
- D- Skull only

**13- The choroid plexus is located in the...?**

- A- Spinal ganglia
- B- Dura mater
- C- Roof of third and fourth ventricles**
- D- Subarachnoid space

**14- The choroid plexus is formed by invagination of the....?**

- A- Ependyma
- B- Pia mater**
- C- Arachnoid mater
- D- Dura mater

**15- Capillaries in the choroid plexus are...?**

- A- Sinusoidal
- B- Fenestrated and dilated**
- C- Non-fenestrated
- D- Continuous

**16- The choroid plexus is covered by...?**

- A- Endothelial cells
- B- Ependymal cells**
- C- Oligodendrocytes
- D- Astrocytes

**17- Ependymal cells are classified as...?**

- A- Columnar cells
- B- Simple cuboidal cells**
- C- Stratified cuboidal cells
- D- Simple squamous cells

**18- A characteristic feature of ependymal cells is the presence of...?**

- A- Myelin sheath
- B- Microvilli and basal folds**
- C- Microvilli only
- D- Cilia only

**19- CSF normally fills all except....?**

- A- Epidural space**
- B- Subarachnoid space
- C- Central canal of spinal cord
- D- Ventricles

**20- Excess CSF drains into venous circulation via...?**

- A- Pia mater
- B- Arachnoid villi**
- C- Dural sinuses directly
- D- Choroid plexus

**21- Arachnoid villi drain CSF into the....?**

- A- Jugular vein
- B- Subclavian vein
- C- Superior sagittal sinus**
- D- Inferior vena cava

**22- Lymphatic vessels in the CNS are...?**

- A- Absent**
- B- Present only in meninges
- C- Few
- D- Abundant

**23- BBB mainly prevents passage of....?**

- A- Water
- B- Toxic substances and some drugs**
- C- Glucose
- D- Oxygen

**24- Capillaries of the blood-brain barrier are....?**

- A- Lined by cuboidal cells
- B- Non-fenestrated with tight junctions**
- C- Sinusoidal
- D- Fenestrated

**25- Astrocyte end-feet are an essential component of the.....?**

- A- Choroid plexus
- B- Brain-CSF barrier only
- C- Blood-brain barrier**
- D- Blood-CSF barrier

**26- The blood CSF barrier includes all except.....?**

- A- Ependymal cells
- B- Astrocyte end feet**
- C- Basement membrane of ependymal cells
- D- Fenestrated endothelium

**27- The brain-CSF barrier consists of....?**

- A- Ependymal cells and astrocyte end-feet**
- B- Pia mater and dura mater
- C- Arachnoid mater and endothelium
- D- Endothelium only

## فورماتيف

### وامتحانات سابقة

**28- What is the structure of the pia mater.....?**

- A- It is formed of web like strands of collagen fibers
- B- It is formed of thick white fibrous connective tissue
- C- It is formed of delicate vascular connective tissue**
- D- It is formed of avascular Fibro-elastic connective tissue

**29- Which of the following pass in the subarachnoid space....?**

- A- Mucus secretion
- B- Blood
- C- CSF**
- D- Lymph

**30- What is the type of the choroidal capillaries.....?**

- A- Blood sinusoids
- B- Fenestrated capillaries**
- C- Non fenestrated capillaries
- D- Continuous capillaries

**31- Which of the following cells share in the formation of blood brain barrier.....?**

- A- Schwann cells
- B- Microglia
- C- Astrocytes**
- D- Ependymal cells

**32- The BBB....?**

- A- Limits the passage of many substances and drugs from the blood to the brain**
- B- Is due to the high permeability of cerebral capillaries
- C- Allows the passage of many substances and drugs from the blood to the brain

**33- The tela choroideae means.....?**

A- Double fold of dura containing the venous blood of the brain

B- Double fold of pia with ependymal cells secretes CSF

C- Double fold of arachnoid with ependymal cells secretes CSF

D- Double fold of dura with ependymal cells secretes CSF

**34- The blood brain barrier is.....?**

A- Astrocyte End-Foot are inside the basement membrane

B- Formed of tight junctions between fenestrated endothelium

C- Contains thick and non-continuous

D- Barrier between blood and CSF and Prevents passage of antibiotics from blood to the nerve tissue

**35- Which of the following cells share in the formation of Blood CSF barrier.....?**

A- Astrocyte

B- Oligodendrocyte

C- Ependymal cell

D- Schwann cell

**36- CSF is absorbed at the...?**

A- Sup. Sagittal sinus

B- Lateral ventricle

C- Third ventricle

D- Inf. Sagittal sinus

**37- ..... Prevents the passage of some substances, such as antibiotics, chemicals and toxic from blood to nerve tissue.**

A- Brain CSF barrier

B- Blood CSF barrier

C- BBB

D- Dura mater

# Histological structure of cerebral and cerebellar cortex

**1- The axon usually arises from the neuron cell body at a region called....?**

- A- Axoplasm
- B- Axon hillock**
- C- Axolemma
- D- Nissl body

**2- Which of the following is absent in the axoplasm.....?**

- A- Nissl bodies**
- B- Microtubules
- C- Neurofilaments
- D- Mitochondria

**3- The plasma membrane of the axon is known as....?**

- A- Endoneurium
- B- Axolemma**
- C- Neurolemma
- D- Axoplasm

**4- Peripheral nerve is best described as.....?**

- A- Only myelinated fibers
- B- Bundles of axons surrounded by connective tissue**
- C- Bundle of dendrites
- D- Single axon

**5- Each individual axon in a peripheral nerve is surrounded by.....?**

- A- Neuroglia
- B- Endoneurium**
- C- Perineurium
- D- Epineurium

**6- The connective tissue sheath that surrounds fascicle of axons is...?**

- A- Dura mater
- B- Perineurium**
- C- Epineurium
- D- Endoneurium

**7- In osmium-stained sections of peripheral nerves myelin appears as....?**

- A- Blue granules
- B- Pink fibers
- C- Black circles**
- D- White rings

**8- Gray matter is mainly composed of...?**

- A- Schwann cells
- B- Connective tissue fibers
- C- Nerve cell bodies, dendrites**
- D- Myelinated axons only

**9- White matter is mainly formed of.....in CNS?**

- A- Astrocytes only
- B- Myelinated axons and oligodendrocytes**
- C- Unmyelinated fibers
- D- Neuron cell bodies

**10- The brain consists of all the following except....?**

- A- Spinal ganglia**
- B- Brain stem
- C- Cerebellum
- D- Cerebrum

**11- The cerebrum is composed of....?**

- A- Brain stem only
- B- Two cerebellar hemispheres
- C- Two cerebral hemispheres**
- D- One hemisphere

**12- The cerebral cortex is arranged into.....?**

- A- Six layers**
- B- Five layers
- C- Four layers
- D- Three layers

**13- The most superficial layer of the cerebral cortex is the.....?**

- A- Inner pyramidal layer
- B- Multiform layer
- C- Molecular layer**
- D- Outer granular layer

**14- The molecular layer mainly contains.....?**

- A- Betz cells
- B- Apical dendrites and nerve fibers of deeper layers**
- C- Granule cells only
- D- Large pyramidal cells

**15- The outer granular layer is mainly formed of.....?**

- A- Purkinje cells
- B- Betz cells
- C- Granule (stellate) cells and neuroglia**
- D- Pyramidal cells

**16- Small and medium pyramidal cells are characteristic of the.....?**

- A- Multiform layer
- B- Molecular layer
- C- Outer pyramidal layer**
- D- Inner granular layer

**17- The inner granular layer is especially thick in.....?**

- A- Cerebellar cortex
- B- Association cortex
- C- Sensory cortex**
- D- Motor cortex

**18- Large pyramidal cells are mainly found in the.....?**

- A- Multiform layer
- B- Molecular layer
- C- Inner pyramidal layer**
- D- Outer pyramidal layer

**19- Betz cells are described as.....?**

- A- Interneurons
- B- Giant pyramidal cells**
- C- Granule cells of cerebellum
- D- Small stellate neurons

**20- Betz cells are located in the.....?**

- A- Cerebellar cortex
- B- Primary motor cortex**
- C- Occipital lobe
- D- Parietal lobe

**21- Axons of Betz cells contribute to the formation of the.....?**

- A- Vestibulospinal tract
- B- Rubrospinal tract
- C- Corticospinal tract**
- D- Spinothalamic tract

**22- The cerebellar cortex is mainly associated with.....?**

- A- Memory
- B- Coordination of muscular activity**
- C- Motor initiation
- D- Sensory perception

**23- The cerebellar cortex consists of.....?**

- A- Six layers
- B- Four layers
- C- Three layers**
- D- Two layers

**24- The outermost layer of the cerebellar cortex is the.....?**

- A- White matter
- B- Molecular layer**
- C- Purkinje cell layer
- D- Granular layer

**25- Purkinje cell bodies form the.....?**

- A- Medullary layer
- B- Purkinje cell layer**
- C- Granular layer
- D- Molecular layer

**26- Purkinje cell dendrites extend into the.....?**

- A- Purkinje layer
- B- Molecular layer**
- C- White matter
- D- Granular layer

**27- Axons of Purkinje cells pass through the granular layer to reach.....?**

- A- Brain stem nuclei
- B- White matter**
- C- Cerebral cortex
- D- Molecular layer

**28- The innermost layer of the cerebellar cortex is the.....?**

- A- White matter
- B- Granular layer**
- C- Purkinje cell layer
- D- Molecular layer

**29- Axons of granule cells are characteristically.....?**

- A- U-shaped
- B- T-shaped**
- C- Spiral
- D- Straight

**30- Golgi cells are located mainly in the.....?**

- A- White matter
- B- Granular layer**
- C- Purkinje cell layer
- D- Molecular layer

**31- Basket and stellate cells are characteristic of the....?**

- A- Cerebral cortex
- B- Purkinje cell layer
- C- Molecular layer**
- D- Granular layer

## فورماتيف

### وامتحانات سابقة

**32- The cerebral cortex is organized into ..... Layers**

- A- Six horizontal**
- B- Three horizontal
- C- Six vertical
- D- Three vertical

**33- In primary motor cortex of frontal lobe .....contains Betz cells**

- A- Outer granular layer
- B- Outer pyramidal layer
- C- Inner granular layer
- D- Inner pyramidal layer**

# Cerebral Circulation and CSF Fluid

**1- CSF examination is done clinically by:**

- A- Ventriculography
- B- CT scan
- C- Lumbar puncture
- D- EEG

**2- All are functions of CSF EXCEPT:**

- A- Shock absorption
- B- Nutrition
- C- Immune defense
- D- Hormone secretion

**3- CSF is absorbed mainly into:**

- A- Arteries
- B- Lymphatics only
- C- Superior sagittal sinus
- D- Ventricles

**4- CSF flows from lateral ventricle to third ventricle through:**

- A- Aqueduct of Sylvius
- B- Foramen of Magendie
- C- Foramen of Monro
- D- Foramen of Luschka

**5- Daily CSF production in adults is:**

- A- 100–200 ml
- B- 200–300 ml
- C- 400–600 ml
- D- 800–1000 ml

**6- CSF is mainly produced by:**

- A- Arachnoid villi
- B- Ependymal cells
- C- Choroid plexus
- D- Pia mater

**7- CSF volume in subarachnoid space is about:**

- A- 100 ml
- B- 25 ml
- C- 125 ml
- D- 50 ml

**8- Total CSF volume in adults is approximately:**

- A- 75 ml
- B- 100 ml
- C- 150 ml
- D- 250 ml

**9- CSF is best described as**

- A- Modified lymph
- B- Ultrafiltrate of plasma
- C- Secretion of neurons
- D- Interstitial fluid

**10- Cerebral ischemia may trigger:**

- A- Babinski reflex
- B- Cushing's reflex
- C- Sheriff reflex
- D- Withdrawal reflex

**11- When ICP rises to 30 mmHg, it leads to:**

- A- Increased CBF
- B- Cerebral hyperemia
- C- Cerebral ischemia
- D- Increased CPP

**12- ICP normally follows the:**

- A- Mean arterial pressure
- B- Cerebral venous pressure
- C- Cerebrospinal fluid pressure
- D- Systolic blood pressure

**13- Increased ICP reduces CBF mainly by:**

- A- Increasing CPP
- B- Decreasing CVR
- C- Compressing cerebral arteries
- D- Increasing arterial pressure

**14- All are mechanisms controlling CVR EXCEPT :**

- A- Metabolic autoregulation
- B- Pressure autoregulation
- C- Chemical control
- D- Hormonal control

**15- Which mechanism allows cerebral vessels to constrict with increased BP and dilate with decreased BP?**

- A- Chemical control
- B- Neural control
- C- Pressure autoregulation
- D- Metabolic control

**16- The range of PaCO<sub>2</sub> over which CBF is highly sensitive is:**

- A- 10–30 mmHg
- B- 20–60 mmHg
- C- 30–80 mmHg
- D- 40–100 mmHg

**17- For each 1 mmHg decrease in PaCO<sub>2</sub>, CBF:**

- A- Increases by 1–2 ml/100 g/min
- B- Decreases by 1 ml/100 g/min
- C- Decreases by 5 ml/100 g/min
- D- Is unchanged

**18- Normal CPP should be above:**

- A- 30 mmHg
- B- 40 mmHg
- C- 50 mmHg
- D- 70 mmHg

**19- Cerebral perfusion pressure (CPP) is defined as:**

- A- MAP + ICP
- B- MAP – ICP
- C- ICP – MAP
- D- CVR × MAP

**20- Neuronal death occurs when CBF falls below:**

- A- 15 ml/100 g/min
- B- 12 ml/100 g/min
- C- 5 ml/100 g/min
- D- 8–10 ml/100 g/min

**21- Cerebral ischemia begins to occur when CBF falls below:**

- A- 30 ml/100 g/min
- B- 25 ml/100 g/min
- C- 18–20 ml/100 g/min
- D- 10 ml/100 g/min

**22- Average cerebral perfusion per 100 g of brain tissue is:**

- A- 20–30 ml/min
- B- 35–40 ml/min
- C- 50–54 ml/min
- D- 70–80 ml/min

## Formatives

### & Previous Exams

**23- Intracranial hypertension is defined when ICP exceeds:**

- A- 15 mmHg
- B- 18 mmHg
- C- 20 mmHg
- D- 25 mmHg

**24- Excessive cerebral blood flow causing raised ICP is called:**

- A- Hypoxia
- B- Ischemia
- C- Hyperemia
- D- Infarction

**25- Normal cerebral blood flow in an adult human is approximately:**

- A- 500 ml/min
- B- 300 ml/min
- C- 750 ml/min
- D- 1200 ml/min

**26- Which of the following causes cerebral vasodilation?**

- A- Decreased  $\text{PaCO}_2$
- B- Increased  $\text{PaCO}_2$
- C- Increased  $\text{PaO}_2$
- D- Decreased hydrogen ion concentration

**27- CBF is best expressed by the equation:**

- A-  $\text{CBF} = \text{CVR} / \text{CPP}$
- B-  $\text{CBF} = \text{CPP} \times \text{CVR}$
- C-  $\text{CBF} = \text{CPP} / \text{CVR}$
- D-  $\text{CBF} = \text{MAP} - \text{ICP}$

**28- The BBB**

- A- Allow the passage of many substances and drugs from the blood to the brain
- B- Is due to high permeability of cerebral capillaries
- C- Limits the passage of many substances and drugs
- D- Is an absolute barrier

**29- Lumbar puncture**

- A- It involves passing a needle into subarachnoid space at the level of L2
- B- Is a sterile procedure to obtain CSF sample
- C- Is safe and has no hazards
- D- Is always not followed by headache

# Basic functions of synapses and neurotransmitters I

**1- The signal of most synapses passes only in forward direction .**

A- True .

B- False .

**2- The nervous system controls the various activities of the body by controlling ..**

A- Contraction of appropriate skeletal ms

B- Contraction of smooth ms in the internal organs

C- Secretion of substances by exocrine and endocrine glands

D- All of the above

**3- Muscle and glands represent ..**

A- Receptors

B- Effectors

C- Centers

D- Afferent neurons

**4- The functional unit of the nervous system**

A- Reflex arc .

B- Brain .

C- Spinal cord .

D- Neurons .

**5- Any change in the environment surrounding a living tissue ....**

A- Receptor

B- Stimulus

C- Center

D- Organ

**6- Present at the peripheral end of the afferent fiber and can detect a stimulus**

A- Stimulus

B- Receptor

C- Afferent neuron

D- Organ

**7- Responsible for sending proper orders to counteract the environmental changes**

A- Stimulus

B- Receptor

C- Center

D- Organ

**8- Communicating neurons may be found in the pathway of which type of reflex arc ?**

A- Monosynaptic

B- Polysynaptic

C- Autonomic synapses

D- B, C

**9- No interneurons can be found in which type of reflex arc ?**

A- Polsynaptic

B- Asynaptic

C- Monosynaptic

D- Autonomic

**10- The most excitable synapse is ..**

A- Axodendritic

B- Axoaxonic

C- Axosomatic

**11- The most common type of synapses ..**

A- Chemical

B- Electrical

**12- The only Bidirectional type of synapses is**

A- Axosomatic

B- Axodendritic

C- Electrical

D- Chemical

**13- Conduction in Electrical synapse is low , And is present in certain areas in brain and cerebellum**

- A- True , False
- B- False , True**
- C- True , True
- D- False , False

**14- Synapse is the site for contact and continuity between pre and post synaptic neurons .**

- A- True
- B- False**

**15- Which of the following is true about synaptic cleft ?**

- A- It is the terminal end of the post synaptic neuron.
- B- Vesicles contains the chemical transmitter.
- C- Space containing ICF separate synaptic knob from the post synaptic membrane.
- D- Space containing ECF separate synaptic knob from the post synaptic membrane.**

**16- The nerve impulse reach the terminal knob after Ca diffusion into the terminal knob.**

- A- True
- B- False**

**17- Which of the following is true about Excitatory synapses (EPSP) ?**

- A- Increase the permeability of post synaptic membrane to Na and K but more to K.
- B- Increase the permeability of post synaptic membrane to Na and K but more to Na.**
- C- Increase the permeability of pre synaptic membrane to Na and K but more to Na.
- D- Hyperpolarization occurs .

**18- Which of the following is true about Inhibitory synapses (IPSP) ?**

- A- GABA and glycine has no role in it .
- B- Increased permeability to Na and proteins .
- C- Depolarization occurs .
- D- K outflux and Cl influx .**

**19- Spontaneous resting flow of Ca is responsible for Miniature postsynaptic potential , few nerve impulses play a role in Miniature postsynaptic potential .**

- A- True . True
- B- True , False**
- C- False , False
- D- False , True

**20- AP in postsynaptic neuron starts at axon hillock because ..**

- A- Its low threshold .
- B- Its high excitability .
- C- Contains many Na channels .
- D- All of the above .**

**21- Repeated stimulation of a single presynaptic fiber at a high rate .**

- A- Temporal summation**
- B- Spatial summation
- C- Lateral inhibition

**22- Stimulation of several excitatory presynaptic neurons that converge on the same post synaptic neuron .**

- A- Temporal summation
- B- Spatial summation**
- C- Lateral inhibition

**23- The inhibitory interneuron in the spinal cord is Renshaw cell , Lateral inhibition has a role in reflex response .**

- A- True , False
- B- Both True**
- C- Both false
- D- False , True

## Formatives

### & Previous Exams

#### 24- Spatial summation

- A- Produce mIPSP
- B- Results from repeated stimulation of a single presynaptic fibers by constant rate
- C- simultaneous stimulation of several excitatory presynaptic neurons converging on same postsynaptic
- D- Can not lead to excitation

#### 25- The synaptic knob contains

- A- Mitochondria only to supply energy
- B- Mitochondria to supply energy for synthesis of and vesicles containing the chemical transmitter.
- C- Receptors of the chemical transmitter.
- D- Enzymes to hydrolyze the chemical transmitter.

#### 26- Excitatory synapses (EPSP):

- A- Is due to the release of an excitatory transmitter as A. Ch. and catecholamines.
- B- Decreases the permeability of postsynaptic membrane to  $\text{Na}^+$  and to  $\text{K}^+$
- C- Is due to the release of GABA.
- D- Produces hyperpolarization of postsynaptic membrane.

#### 27- Miniature postsynaptic potential (mPSP):

- A- Is always inhibitory (mIPSP).
- B- Is always excitatory (mEPSP).
- C- Can be summated to produce an action potential in the postsynaptic cell.
- D- Requires the arrival of an action potential to the nerve terminal.

#### 28- As regards synaptic transmission:

- A- Hypoxia increases synaptic transmission and leads to convulsions
- B- Anesthetics inhibit synaptic transmission via closure of  $\text{K}^+$  channels
- C- It can occur in both directions
- D- Rapid repeated stimulation causes initial potentiation followed by fatigue

# Basic functions of synapses and neurotransmitters II

**1- Ach is secreted by all of these areas except .....**

- A- Postganglionic parasympathetic neurons
- B- Pyramidal cells from motor cortex
- C- All autonomic preganglionic neurons
- D- Most of postganglionic sympathetic neurons**

**2- Norepinephrine is secreted by .....**

- A- Hypothalamus
- B- Locus ceruleus neurons
- C- Medulla oblongata in brain stem
- D- A & B**

**3- Dopamine effect is mainly on .....**

- A- Substantia nigra
- B- Cerebellum
- C- Striatal region of basal ganglia**
- D- None

**4- ..... effect is always inhibitory, while ..... is always excitatory**

- A- Glycine / GABA
- B- GABA / Ach
- C- GABA / Serotonin
- D- Glycine / Glutathione**

**5- Median raphe in brain stem has nuclei secreting .....**

- A- GABA
- B- Serotonin**
- C- NO
- D- Epinephrine

**6- Memory functions may be explained by ..... system**

- A- Serotonin
- B- Ach
- C- NO**
- D- GABA

**7- All of these parts share in formation of Neuropeptides except .....**

- A- Ribosomes
- B- RER
- C- Golgi
- D- SER**

**8- Transporting of Neuropeptides to the tips of nerve fibers is .....**

- A- Very fast
- B- Very slow**

**9- Which of these actions can be caused by Neuropeptides?**

- A- Prolonged opening of Ca channels
- B- affects AP mainly
- C- Prolonged changes in metabolism**
- D- Activation of some genes only

**10- ..... significance is Amplification, while ..... is concentrating of inputs**

- A- Convergence / Divergence
- B- Divergence / Convergence**

**11- Convergence allows .....**

- A- No summation
- B- Temporal summation
- C- Spatial summation**
- D- None

**12- Propriospinal tract is afferent sensory fibers from the dorsal roots to the same level**

- A- True
- B- False**

**13- Closed chain circuit significance is**

- A- +ve feedback
- B- In Facilitatory RF of brain
- C- Continuously stimulates the IC and Ap C
- D- All of the above**

**14- If the synaptic delay is 2 ms , it may be .....**

A- Monosynaptic reflex

**B- Polysynaptic reflex**

**15- Generalized muscle weakness after epileptic fits caused by .....**

A- Chemical transmitters are more than need

**B- Rate of chemical transmitters release > synthesis**

C- Rate of chemical transmitters synthesis > release

D- Weak stimulation of presynaptic knob

**16- Unconsciousness may happen when cerebral circulation is interrupted for**

A- 1S

B- 0.5 ms

**C- 4S**

D- 2S

**17- Hyperventilation may cause ....., while Hypoventilation cause .....**

A- Acidosis / Alkalosis

**B- ↑ excitability / ↓ excitability**

C- ↓ excitability / ↑ excitability

D- Diabetic coma (uncontrolled DM) / Tetanic convulsions

**18- Synaptic transmission is facilitated by ↓ inhibitory transmitters under effect of**

A- Anesthetics

B- Caffeine

**C- Strychnine**

D- None

**19- Which of the following small neurotransmitters is class II ?**

A- NO

**B- Histamine**

C- GABA

D- Ach

**20- The synthesized chemical transmitters in terminal cytosol are transmitted into the vesicle by**

A- Simple diffusion

B- Osmosis

**C- Active transport**

D- Facilitated diffusion

**21- Small neurotransmitters action mainly depends on .....**

A- Changes in cellular metabolism

B- Changes in genes expression

**C- Changed in AP**

D- None

**22- ..... in recycled vesicles are required for concentration of new transmitters into the vesicle**

A- Carbohydrates

B- Fats

**C- Transport Proteins**

D- Vitamins

**23- Cholinesterase is present in .....**

**A- Proteoglycan reticulum terminal**

B- مفيش حاجة تاني رايح فين 🙄

## Formatives

### & Previous Exams

#### 24- Drugs that facilitate synaptic transmission e.g. caffeine

- A-  Transmitter destruction
- B-  Na influx and  K outflux
- C-  the release of inhibitory transmitter
- D-  the release of excitatory transmitter

#### 25- As regards synaptic transmission:

- A- It can occur in both directions
- B- Anesthetics inhibit synaptic transmission via closure of K<sup>+</sup> channels
- C- Hypoxia increases synaptic transmission and leads to convulsions
- D- Rapid repeated stimulation causes initial potentiation followed by fatigue

#### 26- The frequency of discharge of action potentials is:

- A- Directly proportionate to the intensity of the stimulus
- B- Directly proportionate to logarithm the intensity of the stimulus
- C- Inversely proportionate to the intensity of the stimulus
- D- Inversely proportionate to logarithm the intensity of the stimulus

#### 27- Regarding synaptic transmission:

- A- Is depressed by alkalosis
- B- Is increased by acidosis
- C- Repetitive stimulation of a single presynaptic neuron leads to development of action potential in the postsynaptic neuron
- D- Is increased by hypoxia

#### 28- Concerning the slowly acting transmitter:

- A- They are formed in synaptic knob and have a prolonged effect
- B- They are formed in huge amount
- C- They are very potent
- D- They are large and stored in clear vesicles in synaptic knobs

#### 29- Drugs that facilitate Synaptic Transmission (e.g. Caffeine):

- A- Increase transmitter destruction
- B- Increase the release of excitatory transmitter
- C- Increase the release of inhibitory transmitters
- D- Decrease Na<sup>+</sup> influx and increase K<sup>+</sup> outflux

#### 30- Nitric oxide:

- A- It is preformed and stored in vesicles in the presynaptic terminal
- B- It usually alters the membrane potential of the postsynaptic neuron
- C- It has no effects on intracellular metabolic functions
- D- Is secreted by nerve terminals in areas of the brain responsible for long-term behavior and memory

#### 31- Glutamate:

- A- Is the most common excitatory neurotransmitter in the brain
- B- Is synthesized in the cell bodies of neurons
- C- Diffuses out of presynaptic neuron and into the postsynaptic neuron
- D- Is derived from the amino acid tryptophan

**32- With regard to neuropeptides, all the following are correct, except:**

- A- Their effect lasts longer than that of other neurotransmitters
- B- They are synthesized in cell bodies
- C- They migrate down the axon to be stored in terminal knobs
- D- They directly activate second messenger formation without binding to a receptor

**33- Regarding chemical transmitter all are true except:**

- A- Catecholamines and acetylcholine are excitatory transmitters
- B- Serotonin is responsible for presynaptic facilitation
- C- GABA is responsible for pre-synaptic inhibition
- D- Opioid peptides are rapidly acting transmitters

**34- Regarding neuronal pool, which is correct:**

- A- Convergence can lead to amplification of a signal
- B- Divergence can lead to intensification of the signal
- C- Multiple input signals from multiple sources onto single neuron is an example of convergence
- D- Multiple input signals from a single source onto single neuron is an example of convergence

**35- Norepinephrine :**

- A- Always activates excitatory receptors.
- B- Always activates inhibitory receptors.
- C- Is secreted by most postganglionic neurons of the sympathetic nervous
- D- Is secreted by all postganglionic neurons

**36- Acetylcholine is secreted by:**

- A- All the postganglionic neurons of the sympathetic nervous system
- B- All preganglionic neurons of the autonomic nervous system
- C- The adrenal medulla
- D- All neurons in the CNS

# Somatosensory function

**1- Which of the following stimuli doesnot have a sensory receptors ?**

- A- light
- B- pain
- C- X- rays**
- D- sounds

**2- sensory nervous systemis a part of somatosensory system**

- A- True
- B- False**

**3- Chemoreceptors carries information about .....**

- A- pain
- B- pressure changes
- C- chemical changes**

**4- Sensation becomes a conscious experience only when it reaches the.....**

- A- Spinal cord
- B- Thalamus
- C- Cerebral cortex**
- D- Cerebellum

**5- A patient with a tumor compressing the posterior limb of the internal capsule would most likely exhibit which sensory deficit?**

- A- Ipsilateral loss of crude touch from the faceb.
- B- Contralateral loss of all somatic sensations from body and face**
- C- Loss of thermal sensation only from the body
- D- Bilateral loss of proprioception

**6- The pathway carrying pain and temperature sensation is the .....**

- A- Dorsal columns
- B- Lateral spinothalamic tract**
- C- Ventral spinothalamic tract
- D- Corticospinal tract

**7- Pain from structures above the tentorium is referred to which area?**

- A- Occipital region
- B- Neck
- C- Frontal region**
- D- Lower jaw

**8- Removal of a large amount of CSF can cause headache mainly because of .....**

- A- Brain swelling
- B- Downward traction on meninges**
- C- Increase in CSF pressure
- D- Activation of muscle spindles

**9- Which of the followin is considered as Chemoreceptors?**

- A- Osmoreceptors**
- B- Nociceptors
- C- Photoreceptors
- D- Vestibular receptors

**10- Proprioceptors are mainly found in .....**

- A- Skin
- B- joints**
- C- Nasal mucosa
- D- Retina

**11- ..... monitors internal environment**

- A- Exteroceptors
- B- Interoceptors**
- C- Proprioceptors

**12- ..... is Unencapsulated**

- A- Merkel discs
- B- Pacinian corpuscles
- C- root hair plexus
- D- Muscle spindles
- E- A&C**
- F- B&D

**13- Mechanism of Stimulation of Receptors is .....**

- A- Receptor potential- Spread of receptor potential- Action potential.
- B- Receptor potential- Action potential- Spread of receptor potential
- C- Action potential- Receptor potential- Spread of receptor potential
- D- Spread of receptor potential- Receptor potential- Action potential.

**14- Which receptor type adapts very quickly to a constant stimulus?**

- A- Pain receptors
- B- Muscle spindles
- C- Pacinian corpuscles
- D- Nociceptors

**15- The first-order neuron of frontal area of facial sensation is located in the .....**

- A- Substantia gelatinosa
- B- Thalamus
- C- Gasserian ganglion
- D- Motor corte

**16- The Maxillary branch carries sensory impulses from .....**

- A- air sinuses
- B- skin of the cheeks
- C- inside of the cheek
- D- anterior 2/3 rd of the tongue

**17- The ascending branch of the trigeminal sensory root carries .....**

- A- Pain and temperature
- B- Motor impulses
- C- Touch and vibration
- D- Proprioceptiononly

**18- The second-order trigeminal sensory neurons cross the midline and ascend as the .....**

- A- Lateral lemniscus
- B- Medial lemniscus
- C- Trigeminal lemniscus
- D- Corticobulbar tract

**19- The third-order trigeminal sensory neuron cell bodies are in the .....**

- A- Gasserian ganglion
- B- PMVNT
- C- Main sensory nucleus
- D- Spinal nucleus of V

**20- The spinal nucleus of the trigeminal nerve is homologous to which spinal cord structure?**

- A- Clarke's nucleus
- B- substantia gelatinosa
- C- Nucleus dorsalis
- D- Nucleus gracilis

**21- In hemiplegia, the loss of sensation from face is complete**

- A- True
- B- False

**22- A destructive lesion of the trigeminal lemniscus on the right side of the pons produces .....**

- A- Ipsilateral loss of pain and temperature
- B- Contralateral loss of touch, pain, and temperature from the face
- C- Bilateral loss of all facial sensation
- D- Ipsilateral loss of proprioception only

**23- in migraine aura, which event precedes the vasodilation phase of the headache?**

- A- Excessive serotonin release at thalamic nuclei
- B- Sudden vasodilation of cortical arteries
- C- Cortical ischemia secondary to vasospasm
- D- Overactivation of trigeminal nociceptive fibers

**24- Which best explains why intracranial headaches are referred to the scalp despite originating from deep structures?**

A- Pain fibers from the cortex travel directly to scalp cutaneous receptors

**B- Cutaneous and visceral afferents converge on common neurons**

C- Nociceptors in the pia mater synapse in the cerebellum

D- CSF pressure changes activate dermatomal mechanoreceptors

## Formatives

### & Previous Exams

**25- Receptor potential generates**

.....

A- an electrotonic current which is transmitted along sensory fibers to the CNS

B- a nerve impulse at the receptive region of the receptor

C- a state of hyperpolarization of the receptor membrane

**D- a nerve impulse at the spike initiating region of the receptor**

**26- Once initiated, the receptor potential**

.....

A- spreads to a long distance along the sensory nerve fiber

B- amplitude is not related to the strength of the stimulus

C- always generates an action potential from the receptor

**D- stays for a variable period depending on the type of the receptor**

**27- Intracranial headache could result from painful stimuli applied on .....**

A- the dura lining the inner surface of the bones of cranial vault

B- the brain tissue

**C- wall of big intracranial veins**

D- arachnoid mater

**28- Regarding the causes of intracranial headache, select the FALSE .....**

A- decreased CSF pressure

**B- spasm of scalp muscles**

C- distension of big venous sinuses

D- pressure upon tentorium cerebelli

**29- Pain receptors in the skin are typically classified as which of the following?**

- A- Free nerve endings
- B- Same type of receptor that detects position sense
- C- Single class of morphologically specialized receptors
- D- Encapsulated nerve ending

**30- Most sensory receptors.....**

- A- are stimulated by different types of stimuli
- B- are stimulated only by specific stimuli
- C- possess a high threshold for their specific stimuli
- D- only 'b' and 'c' are correct

**31- Pain receptors in the skin are typically classified as which of the following?**

- A- Encapsulated nerve ending
- B- Single class of morphologically specialized receptors
- C- Same type of receptor that detects position sense
- D- Free nerve endings

**32- Slowly adapting receptors include all the following types, except .....**

- A- Golgi tendon organs
- B- warmth receptors
- C- Meissner corpuscles
- D- free nerve endings

**33- The frequency of discharge of action potentials is .....**

- A- Directly proportionate to the intensity of the stimulus
- B- Directly proportionate to logarithm the intensity of the stimulus
- C- Inversely proportionate to the intensity of the stimulus
- D- Inversely proportionate to logarithm the intensity of the stimulus

**38- Trigeminal neuralgia**

A- Loss of sensation in area supplied by trigeminal N. due to a lesion in the gasserian ganglion

B- Corneal ulcer can occur

C- Attacks of severe pain referred in the area of distribution of trigeminal N.

D- Attacks of severe pain referred to base of the neck

**39- As regards the pathway of tactile sensations from the face, the 1st order neuron relays on the .....**

A- Main sensory nucleus of trigeminal nerve in the pons

B- Gracile and cuneate nuclei in the medulla

C- Spinal nucleus of trigeminal nerve in the pons, medulla and upper 2 cervical segments

D- Substantia gelatinosa of Rolandi in the posterior horn of spinal cord

**40- ..... is the conscious awareness & interpretation of a sensation**

- A- Perception
- B- sensation

**41- Sensory receptors are specialized structures at the end sensory neurons that act as**

- A- Detectors for energy changes in CNS
- B- Transducers
- C- Detectors and transducers
- D- None of the above

**42- Meissner's corpuscles**

A- Capsulated and responsible for touch sensation

B- Uncapsulated and responsible for cold sensation

C- Uncapsulated and responsible for touch sensation

D- Capsulated and responsible for hot sensation

# Pain I

## 1- The cutaneous pain:

- A- Is often associated with sweating and lowering of arterial B.P
- B- Is never associated with hyperalgesia
- C- Receptors are free nerve ending**
- D- Is often poorly localized
- E- Is usually referred to other sites

## 2- About hyperalgesia:

- A- It's a physiological condition
- B- It occurs only in association with cutaneous pain
- C- Primary hyperalgesia is due to lesion in the CNS
- D- The pain threshold is lowerd in primary hyperalgesia**

## 3- About the fast cutaneous pain , all of the following is true except:

- A- It's conducted along A delta nerve fibers
- B- It initiates the protective withdrawal reflex
- C- It's rapidly abolished by local anesthetic drugs**
- D- It's transmitted to the cerebral cortex via the lateral spinothalamic tract

## 4- Which stimuli produces immobilization reflex ?

- A- Mechanical
- B- Thermal
- C- Chemical**

## 5- Type of pain which is due to spasm of hollow viscrea

- A- Pricking pain
- B- Burning pain
- C- Aching pain
- D- Colicky pain**

## 6- The center of immediate pain is

- A- Cerebral cortex**
- B- The thalamus
- C- Spinal cord
- D- Non

## 7- All of the following are characteristic for delayed pain except:

- A- Diffuse burning
- B- Can summate and becomes intolerable
- C- Its rate of conduction is 0.5 to 2 m/s
- D- Affected by local anaethetics slower than immediate pain**

## 8- Help localizing the site of pain sharply

- A- Thalamus
- B- Dorsal root ganglia
- C- Cerebral cortex**
- D- Reticular formation

## 9- Axons of PLVNT ascends in the posterior limb of the IC to form the sensory radiation to end in the sensory area of the CC in the precentral gyrus

- A- ☒
- B- ☐

## 10- All of the following are from characters of immobilization reflex except:

- A- Protective
- B- Proponent
- C- Can inhibit all other reflexes at the same time
- D- Its center is cerebral cortex**

## 11- All individuals have the same behavior to pain, but the differ in their threshold for pain

- A- ☒
- B- ☐

**12- Severe pain causes**

- A- Increased Blood pressure
- B- Tachycardia
- C- Vomiting**
- D- Non

## Formatives

### & Previous Exams

**13- Free nerve ending are receptors of**

- A- Fine touch
- B- Vibration
- C- Pressure
- D- Pain**

**14- Which statement is correct:**

- A-  $A\beta$  fibres are thickly myelinated and normally transmit nociceptive stimuli
- B- A  $\delta$  fibers are unmyelinated and slower conducting than  $A\beta$
- C- A  $\delta$  fibers can alter the rate of firing depending on the intensity of stimuli**
- D- C fibers are not myelinated and rapid conducting

**15- Fast pain is transmitted by :**

- A-  $A\beta$  fibers
- B- A  $\delta$  fibers**
- C- C fibers
- D- Motor fibers

**16- Delayed pain is transmitted by**

- A- A  $\beta$  fibers
- B- A  $\delta$  fibers
- C- C fibers**
- D- Motor fibers

**17- Secondary hyperalgesia:**

- A- Its mechanism may be due to skin damage and release of PPS
- B- The pain threshold is lowered
- C- Occurs in healthy skin around the injured area**
- D- Occurs in all the sites of skin area showing redness (red flare)

**18- Pain receptors in the skin are typically classified as which of the following?**

- A- Encapsulated nerve ending
- B- Single class of morphologically specialized receptors
- C- Same type of receptor that detects position sense
- D- Free nerve endings**

**19- The pain impulses serving emotional reactions should be relayed in GINGULATE GYRUS**

- A- ✓**
- B- ✗

**20-Primary hyperalgesia :**

- A- Occurs in healthy skin around the injured area
- B- Occurs in all sites of skin area showing redness (red flare)**
- C- The pain threshold is not lowered
- D- Its mechanism may be due to convergence-facilitation or lateral inhibition

# Pain II

## 1- About visceral pain, all the following is true EXCEPT:

- A- Is poorly localized
- B- May cause Reflex contraction of the overlaying muscles
- C- Is accompanied by pressor effects as increased ABP**
- D- Is carried on C nerve fibers

## 2- About referred pain, all the following is true EXCEPT:

- A- It occurs as a result of migration of organ during development
- B- It is explained by the convergence-projection theory
- C- It always accompanies cutaneous pain.**
- D- It is a major manifestation of visceral pain.

## 3- Gallbladder pain is referred to the:

- A- Umblicus
- B- Tip of the right scapula**
- C- Left shoulder
- D- Left lumbar region

## 4- The visceral pain receptors:

- A- Are more abundant than the cutaneous pain receptors.
- B- Are rapidly-adapting receptors.
- C- May discharge in somatic nerves.**
- D- Discharge impulses in thick afferent myelinated nerve fibres.

## 5- The visceral pain:

- A- Is commonly localized with spasm of its overlaying muscles.
- B- Can be elicited by either distension or traction of viscera.**
- C- Transmitted by somatic nerve fibres only

## 6- Visceral pain may result from the following EXCEPT:

- A- Ischaemia
- B- Rupture gastric peptic ulcer.
- C- Overdistension of hollow viscera.
- D- Cutting through the viscera.**

## 7- Deep pain:

- A- Is due to stimulation of pain receptors in the skin.
- B- Is pricking in character
- C- Is poorly localized.**
- D- Is accompanied by tachycardia and hypertension.

## 8- Concerning visceral pain, following are true statements, EXCEPT:

- A- May cause reflex vomiting.
- B- May cause contraction of overlying skeletal muscle.
- C- Is well localized.**
- D- May cause change in arterial blood pressure

## 9- Pain arising from the viscera:

- A- May be transmitted by somatic nerve fibres.
- B- May lead to reflex contraction of the nearby skeletal muscles.
- C- Is poorly-localized compared to cutaneous pain.
- D- All of the above.**

## 10- About ischaemic pain, all the following is true EXCEPT:

- A- It often occurs with deep and visceral pain.
- B- It occurs in all types of muscle (skeletal, smooth and cardiac muscles).
- C- Angina pectoris is a classical example.
- D- It is due to release of K<sup>+</sup>, kinin or lactic acid.
- E- In case of intermittent claudication, it occurs mainly during rest**

**11- The visceral pain receptors:**

A- Are more abundant than the cutaneous pain receptors.

B- Are rapidly-adapting receptors.

**C- May discharge in somatic nerves.**

D- Initiate pain that reaches the cerebral cortex via the gracile and cuneate tracts.

E- Discharge impulses in thick afferent myelinated nerve fibres

**12- About stimulation of pain receptors in hollow viscera, all the following is true EXCEPT:**

A- They are often stimulated as a result of ischaemia.

**B- They may be stimulated by cutting through the wall with a sharp scalpel.**

C- They can be stimulated by distension of the viscera.

D- Inflammation of the visceral wall often stimulates them.

E- They are stimulated when strong rhythmic contractions occur in the wall of an obstructed viscus

**13- Severe itching may be because of**

A- Renal failure

B- HIV infection

C- Atopic dermatitis

**D- All of the above**

## Formatives

### & Previous Exams

**14- Deep pain is characterized by all of the following except:**

A- It is dull aching

B- It's conducted by C nonmyelinated

**C- It's sharp localized pickling**

**15- Referred pain includes all the following EXCEPT:**

A- Cardiac pain.

**B- Muscle pain.**

C- Gastric pain.

D- Renal pain.

**16- Visceral pain is characterized by all the following EXCEPT:**

A- It is dull aching or colicky.

**B- Transmitted by A  $\delta$  fibers.**

C- It is diffuse and not localized.

D- It is usually referred.

**17- 1. Deep pain is characterized by all the following EXCEPT:**

A- It is dull aching.

B- It is conducted by C nonmyelinated.

**C- It is sharp localized pricking.**

D- It is associated by reflex immobilization.

**18- Opiate receptors are present in all these sites EXCEPT:**

A- Periventricular neurons.

B- Priaqueductal neurons,

C- Substantia Nigra

**D- AHCs**

**19- Evidences supporting the gate theory for pain control are the following EXCEPT:**

A- Stress analgesia.

**B- Reflex rigidity (immobilization).**

C- Rubbing of the skin around the injured area decrease the degree of pain sensation.

D- Morphine is used as a powerful analgesic drug to reduce pain sensation.

**20- Pain from the stomach is referred to which area of the body**

A- Upper right shoulder

B- Proximal of the anterior and inner thigh

**C- Abdominal area above the umbilicus**

D- Abdominal area below the umbilicus

# Stretch Reflex I & II

## 1- Inverse stretch reflex:

A- increases the possibility of avulsion of the excessively stretched muscle from its bony attachments

B- has no reciprocal innervation circuits

C- is clinically manifested by lengthening reaction

D- is clinically tested by examining the tendon jerks

## 2- Skeletal muscle tone:-

A- is a dynamic stretch reflex

B- has a dynamic state

C- is increased during rest

D- is decreased during standing upright

## 3- Adequate level of muscle tone is essential for the accurate performance of voluntary movements because

A- it adjusts the a-motor neuron discharge initiating voluntary movements

B- it adjusts posture of proximal joints of the acting limbs

C- it adjusts the upright posture of the whole body during voluntary movements

D- it adjusts sensitivity of the spindles of the involved muscles

## 4- A tendon jerk

A- is a dynamic stretch reflex

B- is a static stretch reflex

C- is evoked by gradually stretching the muscle

D- is evoked by stimulation of tendon receptors

## 5- Skeletal muscle tone:

A- is a voluntary act.

B- keeps posture against gravity.

C- increases if the muscle shortens.

D- all of the above are correct.

E- only a and b are correct.

## 6- Dynamic stretch reflex is:

A- slower than withdrawal reflex.

B- the mechanism of muscle tone.

C- polysynaptic reflex.

D- increased in UMNL.

E- increased in neocerebellar lesion.

## 7- Primary ending from muscle spindle:

A- discharge when muscle stretch is maintained.

B- explain alpha – gamma coactivation.

C- is the afferent for dynamic stretch reflex.

D- is the afferent for static stretch reflex.

E- is the afferent for muscle tone.

## 8- Static stretch reflex is

A- the base of tendon jerk.

B- stimulated by red nucleus.

C- a polysynaptic reflex.

D- inhibited by vestibular nucleus.

E- the base of muscle tone.

## 9- Static stretch reflex is:

A- responsible for tendon jerk.

B- transmitted through primary ending only.

C- initiated at nuclear chain intrafusal fibers.

D- initiated at nuclear bag intrafusal fibers.

E- initiated by neocerebellum.

## 10- Dynamic stretch reflex:

A- has secondary ending as an afferent

B- has alpha neuron as an afferent.

C- the mechanism of muscle tone.

D- has gamma motor neuron as an efferent.

E- a & c are correct

**11- The dynamic stretch reflex:**

- A- is the basis of muscle tone.
- B- its receptor is nuclear chain.
- C- is mediated through stimulation of the secondary nerve endings.
- D- shows rapid contraction of the muscle followed by rapid relaxation.**
- E- is maintained as long as the stretching of the muscle is maintained

**12- Gamma motor neuron to muscle spindle**

- A- supplies the central receptor areas of intrafusal muscle fibers.
- B- supplies the extrafusal muscle fibers.
- C- constitutes the pathway through which higher centers can affect the stretch reflex.**
- D- constitutes the efferent for stretch reflex.
- E- constitutes the afferent for stretch reflex

**13- Muscle tone:**

- A- is less in antigravity muscles
- B- decreases venous return.
- C- is based on the static stretch reflex**
- D- is characterized by fatigue.
- E- is based on the dynamic stretch reflex

**14- Muscle tone:**

- A- is the dynamic stretch reflex.
- B- is due to discharge in the primary ending of muscle spindle.
- C- is less prominent in antigravity muscles.
- D- is increased in LMNL.
- E- none of the above**

**15- Muscle spindle:**

- A- is present in muscle tendon
- B- has afferent A-delta fibers.
- C- has afferent C fibers.
- E- is facilitated by motor area 4.**

**16- Nuclear bag fibers in muscle spindle are:**

- A- slowly responding.
- B- slowly adapting.
- C- rapidly responding.**
- D- connected to secondary ending.
- E- responsible for static stretch reflex

**17- The tendon jerk which has its center in the 5th and 6th cervical segments of the spinal cord is**

- A- the jaw jerk
- B- the deltoid jerk
- C- the biceps jerk
- D- the triceps jerk**

**18- Tendon jerks are clinically examined to assess**

- A- integrity of muscle spindles
- B- integrity of reflex pathway**
- C- the total reflex time of the jerk
- D- central delay time of the jerk

**19- Absence of a tendon jerk could result from any of the following conditions, except:**

- A- lesions of supraspinal facilitatory centers**
- B- lesions of the efferent neurons
- C- lesions of the afferent neurons
- D- lesions of the spinal nerve centers

**20- Exaggeration of tendon jerks could result from any of the following conditions, except**

- A- lesions of supraspinal facilitatory centers**
- B- lesions of supraspinal inhibitory centers
- C- increased gamma motor neuron discharge
- D- anxiety

**21- When a skeletal muscle is suddenly stretched:**

- A- it relaxes suddenly
- B- it develops a static stretch
- C- it develops a dynamic stretch reflex**
- D- it develops clonic contractions

**22- Anti-gravity muscles maintain stretch reflex for prolonged periods without fatigue, because**

A- they are heavily innervated by a motor neurons

B- they obtain their energy needs mainly from anaerobic metabolic processes

C- they contain exceptionally high levels of creatine phosphate

D- they are rich in mitochondria

**23- The discharge from Golgi tendon organs initiated by excessive stretch of a skeletal muscle produces**

A- inhibition of alpha motor neurons of antagonistic muscles

B- inhibition of gamma motor neurons of antagonistic muscles

C- inhibition of alpha motor neurons of the same muscle

D- stimulation of gamma motor neurons of the same muscle

**24- The center cremasteric reflex**

A- T9-T10

B- S1-S2

C- L1-L2

D- T6

**25- The center of planter Reflex**

A- T10

B- L1-L2

C- T1-T2

D- S1-S2

**26- The center of abdominal Reflexes at the level of umbilicus**

A- T7-T8-T9

B- T10

C- T11-T12

D- S1-S2

**27- The center of Knee clonus is**

A- T10

B- L1-L2

C- L3-L4

D- S1-S2

**28- A typical example of a monosynaptic reflex is:**

A- stretch reflex.

B- superficial abdominal reflex.

C- withdrawal reflex.

D- light reflex.

**29- The crossed extensor reflex is best described by:**

A- Flexion of both lower limbs

B- Extension of the stimulated limb only

C- Flexion of the ipsilateral limb with extension of the contralateral limb

D- Extension of both limbs simultaneously

**30- In the stepping reflex of a single limb:**

A- Flexion is followed by flexion

B- Extension is followed by extension

C- Forward flexion is followed by backward extension

D- No rhythmic movement occurs

**31- Reciprocal stepping reflex of the opposite limb means:**

A- Both limbs flex together

B- Both limbs extend together

C- Flexion of one limb with extension of the opposite limb

D- Random limb movement

**32- Diagonal stepping of all four limbs is characterized by:**

A- Flexion of both forelimbs together

B- Flexion of both hindlimbs together

C- Flexion of right forelimb with left hindlimb

D- Extension of all limbs simultaneously

**33- All the following are true about the flexion withdrawal reflex EXCEPT:**

- A- It is a polysynaptic spinal cord reflex
- B- It has a central delay more than 2 ms
- C- It shows reciprocal inhibition
- D- It is a monosynaptic stretch reflex

**34- Withdrawal reflex means that it:**

- A- Is easily fatigued compared to other reflexes
- B- Has a short central delay
- C- Inhibits other reflexes occurring at the same time
- D- Is mediated only by Ia afferent fibers

## Formatives

### & Previous Exams

**35- Clonus occurs when:**

- A- Tapping on the tendon of the muscle in normal person.
- B- Tapping on the tendon of the muscle in hypotonia.
- C- Tapping on the tendon of the muscle in hypertonia.
- D- There is maintained stretch on muscle tendon in cases of hypertonia.

**36- The center for the knee jerk is:**

- A- C4–C5
- B- S1–S2
- C- L3–L4
- D- C5–C6

**37- The tendon jerk is lost (areflexia) if**

- A- The reflex arc is injured at any facilitatory point.
- B- The supraspinal facilitatory centers or their descending tracts are affected.
- C- The supraspinal inhibitory centers or their descending tracts are affected.
- D- Increased muscle tone

**38- The gamma (efferent) motor fibers:**

- A- Supply the central part of the intrafusal muscle fibers.
- B- Supply the extrafusal muscle fibers.
- C- Are axons of the large AHCs ( $\alpha$  AHCs).
- D- Supply the peripheral contractile part of the intrafusal muscle fibers.

**39- Maximal stimulation and discharge of the muscle spindle occurs when:**

A- Muscle contraction occurs without ↑ gamma efferent discharge.

B- The muscle is stretched and there is ↑ gamma efferent discharge.

C- There is ↑ gamma efferent discharge.

D- Tapping on the muscle tendon with a hammer.

**40- The dynamic response of the stretch reflex is characterized by the following EXCEPT:**

A- Its afferents are the Iry endings.

B- The response is maintained subtetanic contraction.

C- The stimulus is sudden stretch.

D- The receptor is nuclear bag fibers

**41- The supraspinal facilitatory centers controlling stretch reflex are the following EXCEPT:**

A- The vestibular nuclei.

B- The facilitatory reticular formation.

C- The neocerebellum.

D- The basal ganglia.

# Protozoa causing CNS diseases I

**1- In which of the following parasite cat is the definitive host?**

- A- Plasmodium vivax
- B- Leishmania
- C- Toxoplasma gondii**
- D- Trypanosoma

**2- Which animal serves as the definitive host for Toxoplasma gondii?**

- A- Birds and mammals
- B- Dogs and foxes
- C- Cats and other felines**
- D- Humans

**3- What is the habitat of T. gondii in definitive host?**

- A- Reticulo-endothelial cells
- B- Enterocytes**
- C- Lymphoid cells
- D- Neuronal cells

**4- Which stage of T. gondii is the rapidly multiplying form observed during the acute phase of infection?**

- A- Bradyzoite
- B- Tachyzoite**
- C- Sporozoite
- D- Oocyst

**5- Concerning Toxoplasma gondii infection takes place through:**

- A- Ingestion of soil contaminated by unsporulated oocysts from cats
- B- Ingestion of cysts in undercooked meat**
- C- Congenital transmission from a mother with chronic toxoplasmosis
- D- Bite of an infected sandfly

**6- The effect of infection by toxoplasmosis on the fetus depends on:**

- A- Time of infection during pregnancy
- B- Immunity of the mother
- C- Time of infection during pregnancy and Immunity of the mother**

**7- Lymphadenitis in toxoplasmosis is associated with a clinical presentation similar to:**

- A- Glandular fever**
- B- Typhoid fever
- C- Measles
- D- Mumps

**8- Which symptom is considered a severe and potentially fatal form of symptomatic chronic Toxoplasmosis?**

- A- Lymphadenitis
- B- Myalgia
- C- Retinochoroiditis
- D- Meningoencephaliti**

**9- The most important and common manifestation of Toxoplasmosis in immunocompromised patients is ....**

- A- Lymphadenitis
- B- Myocarditis
- C- Retinochoroiditis
- D- Encephalitis**

**10- Acute disseminated Toxoplasmosis is primarily observed in patients receiving:**

- A- Routine antibiotics
- B- Chemotherapy for cancer
- C- Organ transplant**
- D- Anti viral therapy

**11- Used for pregnancy woman to reduce the risk of transplacental transmission:**

- A- Atovaquone
- B- Corticosteroids
- C- Spiramycin**
- D- Flutamid

**12- Which form of *Toxoplasma gondii* is observed during the chronic phase of infection?**

- A- Tachyzoite
- B- Bradyzoite**
- C- Sporozoite
- D- Trophozoite

**13- Which diagnostic method is considered the "Gold Standard" for serological testing of *T. gondii*?**

- A- ELISA (IgM/IgG)
- B- PCR
- C- Sabin Feldman dye test (IgG)**
- D- Indirect Hemagglutination Test (IHAT)

**14- Which imaging finding often reveals cerebral lesions such as calcification and hydrocephalus?**

- A- PCR
- B- Ultrasonography
- C- CT, MRI**
- D- echo

**15- Which combination of drugs is the standard treatment regimen for acute symptomatic Toxoplasmosis?**

- A- Spiramycin and Atovaquone
- B- Sulfadiazine and Atovaquone
- C- Pyrimethamine and Sulfadiazine**
- D- Corticosteroids and Pyrimethamine

**16- Which type of infection typically requires no treatment?**

- A- Congenital infection
- B- Acute symptomatic infection
- C- Chronic or latent infection**
- D- Encephalitis in AIDS patient

**17- Used in the treatment of ocular toxoplasmosis primarily to Decrease the inflammatory reaction**

- A- Corticosteroids**
- B- Pyrimethamine
- C- Spiramycin
- D- Atovaquone

**18- In Sabin Feldman dye test in positive cases trophozoites are**

- A- Stained blue
- B- Not attached to antibodies
- C- Not stained**
- D- None of them

**19- Late in congenital infection of *T.gondii* the infection**

- A- Become generalized
- B- Localized in viscera
- C- Clears from all body
- D- Localized mainly in CNS**

**20- How to prevent and control infection of *T.gondii***

- A- Proper washing of hands
- B- Safe disposal of cat's feces
- C- Freezing of meat to (-20c)for 2 days may kill the cyst
- D- All are right**

**21- Which symptom is considered a severe and fatal with symptomatic chronic Toxoplasmosis?**

- A- Myocarditis**
- B- Lymphadenitis
- C- Chorioretinitis
- D- Splenomegaly

**22- ..... is used for encephalitis in AIDS**

- A- Corticosteroids
- B- Atovaquone**
- C- pyrimethamine
- D- Spiramycin

**23- ..... cause PAM**

- A- African Trypanosoma
- B- *Toxoplasma gondii*
- C- *Naegleria fowleri***
- D- Acanthamoeba



**38- Granulomatous amoebic encephalitis (GAE) is rarely diagnosed before death. Most cases have been diagnosed post-mortem or shortly before death.**

A- 

B- 

**39- Treatment of Acanthamoeba**

A- Excision of focal lesion + Ketoconazole

B- Penicillin

C- chloramphenicol

D- All of above

**40- Naegleria Meningoencephalitis is ..... But Acanthamoeba Encephalitis is .....**

A- Diffuse/focal

B- يا رب نكسب امم افريقيا 🇪🇬🏆👓

**41- Amoebic brain abscesses caused by.....**

A- Entamoeba histolytica

B- Toxoplasma gondii

C- African Trypanosoma

D- American Trypanosoma

**42- Infective stage of Entamoeba histolytica to the CNS**

A- Quadri-nucleated cyst

B- Trophozoite

C- Both

D- Non

**43- Amoebic brain abscesses may be misdiagnosed to brain tumour**

A- 

B- 

**44- Treatment for Amoebic brain abscesses is .....**

A- Emetine only

B- Metronidazole only

C- Combination of the two drugs

## Formatives

### & Previous Exams

**45- Amphotericin B used in the treatment of**

A- Leishmania spp.

B- Entamoeba histolytica

C- Trypanosoma brucei

D- Naegleria fowleri

**46- Avoidance of swimming or bathing in contaminated water is essential for prevention and control of**

A- African sleeping sickness

B- Brain amoebic abscess

C- Malignant falciparum malaria

D- Primary amoebic meningoencephalitis

**47- The diagnostic stage of granulomatous amoebic encephalitis is (G.A.E):**

A- Trophozoite stage.

B- Cyst stage.

C- Trophozoite and cyst

**48- The infective stage in case of Negleriasis is:**

A- Trophozoite

B- Cyst

C- Trophozoite and cyst

**49- Naegleria fowleri infection is acquired by:**

A- Swimming in fresh water

B- Drinking of contaminated water

C- Walking bare footed on moist sand soil

D- Blood transfusion

**50- Which parasitic stage is responsible for amoebic brain abscess?**

A- Trophozoite

B- Uninucleate cyst

C- Binucleate cyst

D- Quadri-nucleate cyst

# Brain metabolism

**1- The main source of energy for the brain under normal conditions is:**

- A- Fatty acids
- B- Amino acids
- C- Glucose**
- D- Ketone bodies

**2- During prolonged starvation, the brain adapts to use:**

- A- Free fatty acids
- B- Amino acids
- C- Lactate
- D- Ketone bodies**

**3- Brain glycogen stores are:**

- A- Absent
- B- Large and long-lasting
- C- Present in large amount in neurons
- D- Very small and mainly in astrocytes**

**4- The main cell involved in brain carbohydrate metabolism is:**

- A- Neuron
- B- Oligodendrocyte
- C- Astrocyte**
- D- Microglia

**5- Inside neurons, glucose is metabolized mainly by:**

- A- Glycogenolysis
- B- Glycolysis and Krebs' cycle**
- C- Pentose phosphate pathway only
- D-  $\beta$ -oxidation

**6- The pentose phosphate pathway in brain is important mainly for production of:**

- A- ATP
- B- NADH
- C- NADPH**
- D- FADH

**7- During the first few days of starvation, the brain depends mainly on:**

- A- Ketone bodies
- B- Glucose derived from muscle protein**
- C- Fatty acids
- D- Lactate

**8- After about 2–3 weeks of starvation, brain utilization of ketone bodies results in:**

- A- Increased muscle protein breakdown
- B- Increased glucose oxidation
- C- Decreased muscle protein breakdown**
- D- Complete cessation of gluconeogenesis

**9- The three amino acids with the highest concentration in brain tissue are:**

- A- Alanine, glycine, serine
- B- Glutamic acid, glutamine, aspartic acid**
- C- Valine, leucine, isoleucine
- D- Tyrosine, tryptophan, phenylalanine

**10- Glutamic acid metabolism in the brain is important mainly for:**

- A- Fat synthesis
- B- Regulation of brain functions**
- C- Glycogen storage
- D- Ketone body synthesis

**11- Red blood cells differ from brain cells in that RBCs:**

- A- Perform Krebs' cycle
- B- Use aerobic glycolysis
- C- End glycolysis by lactic acid formation**
- D- Can use ketone bodies

**12- In the fed state, excess glucose in the liver is mainly converted into:**

- A- Ketone bodies
- B- Lactate
- C- Glycogen and TAG**
- D- Amino acids

**13- In muscle tissue, excess glucose is stored as:**

- A- TAG
- B- Glucose-6-phosphate
- C- Glycogen**
- D- Lactate

**14- The brain requires a continuous supply of glucose because it:**

- A- Cannot synthesize ATP
- B- Has minimal energy stores**
- C- Uses anaerobic metabolism only
- D- Cannot use oxygen

**15- Energy produced per gram of fat is approximately:**

- A- 4 kcal
- B- 7 kcal
- C- 9 kcal**
- D- 12 kcal

**16- Energy produced per gram of carbohydrate is approximately:**

- A- 2 kcal
- B- 4 kcal**
- C- 7 kcal
- D- 9 kcal

**17- Gluconeogenesis in brain tissue produces:**

- A- Free glucose
- B- Glucose-1-phosphate
- C- Glucose-6-phosphate**
- D- Glycogen

## Formatives

### & Previous Exams

**18- The main source of energy for brain**

- A- Fatty acid
- B- Glycin
- C- Glucose**
- D- Keton bodies

**19- AA. With the highest levels of in the brain tissue**

- A- Alanine
- B- Glycin
- C- Leucine
- D- Aspartate**

**20- Fuel used by brain is:**

- A- Glucose**
- B- Glycerol
- C- Lactic acid
- D- Propionic acid

**21- Amino acid with the highest levels in brain tissue is:**

- A- Alanine
- B- Glutamic acid**
- C- Glycine
- D- Leucine

**22- After three weeks of starvation, the energy of the brain is derived from oxidation of:**

- A- Proteins
- B- Ketone bodies**
- C- Glucose
- D- Glycogen

**23- Energy derived from carbohydrate oxidation is:**

- A- 4 Kcal**
- B- 9 KCal
- C- 10 KCal
- D- 12 KCal

**24- During first few days of starvation, brain uses glucose derived from:**

- A- Muscle protein
- B- Free fatty acids
- C- Ketone bodies
- D- Fructose

**25- The brain depends firstly on \_\_ as a source of energy:**

- A- Glucose
- B- Free fatty acids
- C- Ketone bodies
- D- Fructose

# CNS Infection

**1- The infective agent may reach the CNS by various routes EXCEPT:**

- A- Direct spread from middle ear
- B- Penetrating wounds
- C- Lymphatic drainage**
- D- Blood stream

**2- in CSF analysis the glucose  $<0.4$  this is characteristic of a**

- A- Normal CSF sample
- B- Turbid CSF sample**
- C- Clear CSF sample
- D- CSF with Fibrin web

**3- The form of meningitis considered the least damaging and self-limited is**

- A- Tuberculosis meningitis
- B- Fungal meningitis
- C- Bacterial meningitis
- D- Viral meningitis**

**4- The classic triad of symptoms for meningitis includes sudden high fever, altered mental status and**

- A- Positive Kernig's sign
- B- Nuchal rigidity**
- C- Arterial thrombosis
- D- Vomiting

**5- A finding of low glucose level ( $<0.4$ ) and a lymphocyte cell predominance suggests**

- A- Normal CSF findings
- B- Acute bacterial meningitis
- C- TB Meningitis**
- D- Viral Meningitis

**6- Negative gram stain in a CSF sample is typically observed in cases of**

- A- Streptococcus pneumoniae meningitis
- B- Turbid CSF samples
- C- Clear CSF samples**
- D- Haemophilus influenzae meningitis

**7- The most common cause of meningitis in neonates is**

- A- Hemophilus influenza
- B- Neisseria meningitidis
- C- E. coli**
- D- Streptococcus pneumoniae

**8- The predominant cell in CSF in viral meningitis is**

- A- Neutrophils
- B- Macrophages
- C- Plasma cells
- D- Lymphocytes**

**9- The classic triad of meningitis is**

- A- Fever, vomiting and confusion
- B- Headache, Fever and Neck stiffness**
- C- Headache, nausea and vomiting
- D- Altered mental, fever and vomiting

**10- Coming home from a summer camp, a young boy presents with fever, headache and a stiff neck. A spinal tap reveals high protein and low sugar in CSF. These findings suggest**

- A- Epidural hematoma
- B- Acute bacterial meningitis**
- C- Epidural abscess
- D- Viral meningitis

**11- Bacterial meningitis cause**

- A- Decrease in level of CSF glucose**
- B- Decrease in cell count
- C- Increase of cell count specially lymphocytes
- D- Clear in appearance

**12- Infection of the brain parenchyma is called**

- A- Meningitis
- B- Encephalitis**
- C- Brain edema
- D- Astrocytoma

**13- Bacteria can reach the brain through**

- A- Direct spread
- B- Lymphatic spread
- C- Transperitoneal spread
- D- Trans pleural spread

**14- Brain abscess is occurred due to**

- A- ear infection
- B- infection of sinuses
- C- severe head injury
- D- all of the above

**15- Regarding Cerebral abscess**

- A- spread mainly by lymphatic spread
- B- Caused by aerobic organism
- C- Can initiate seizures
- D- It has a specific microscopic picture

**16- Viral meningitis**

- A- It is a fatal condition
- B- caused mainly by enteroviruses
- C- Lead to severe neurological dysfunction
- D- There is of monocytes in the CSF

**17- Which is characteristic for encephalitis**

- A- CMV is the commonest cause
- B- Common condition
- C- Herpes simplex cause severe encephalitis
- D- It is self limited disease

**18- A 22 years old woman was admitted to the hospital with a 1 day of high fever, headache and neck stiffness. CSF test result was positive with gram negative diplococci. What is the causative organism of this case**

- A- Pneumococci
- B- Neisseria meningitides
- C- Staph. aureus
- D- Listeria

**19- Meningitis due to Streptococcus pneumoniae is primarily noted for occurring in**

- A- Young adults and adolescents
- B- Debilitated and elderly persons or young children
- C- Immunocompetent patients
- D- All patients equally due to widespread infection

**20- A cerebral abscess most commonly spreads from which source**

- A- Skin ulcers
- B- Middle ear or mastoid infection
- C- Lung carcinoma
- D- Renal infection

# Antidepressant drugs

**1- Depression is mainly associated with deficiency of which neurotransmitters?**

- A- GABA & Glutamate
- B- Dopamine only
- C- Norepinephrine & Serotonin**
- D- Acetylcholine

**2- Which theory explains depression as a deficiency of monoamines?**

- A- Neurogenic theory
- B- Biogenic amine theory**
- C- Glutamatergic theory
- D- Dopaminergic theory

**3- Which of the following is an SSRI?**

- A- Venlafaxine
- B- Imipramine
- C- Fluoxetine**
- D- Bupropion

**4- Which SSRIs are potent CYP2D6 inhibitors?**

- A- Fluoxetine & Fluvoxamine
- B- Sertraline & Escitalopram
- C- Fluoxetine & Paroxetine**
- D- Citalopram & Sertraline

**5- Which SSRI may cause QT prolongation?**

- A- Paroxetine
- B- Fluoxetine
- C- Escitalopram
- D- Citalopram**

**6- Discontinuation syndrome is least likely with which SSRIs:**

- A- Fluoxetine**
- B- Sertraline
- C- Citalopram
- D- Paroxetine

**7- Right about SSRIs**

- A- Block reuptake of Norepinephrine
- B- Produce improvement in mood in 5 days at least
- C- Citalopram used in bulimia
- D- Lead to increased concentration of the neurotransmitter in the synaptic cleft**

**8- Sertraline differ from other in**

- A- Longer half life
- B- Approved Using in bulimia
- C- Cause QT prolongation
- D- Food increase its absorption**

**9- SNRIs inhibit reuptake of:**

- A- Serotonin only
- B- Norepinephrine only
- C- Both serotonin and norepinephrine**
- D- Dopamine

**10- Venlafaxine at low doses is a potent inhibitor of :**

- A- Dopamine reuptake
- B- Norepinephrine reuptake
- C- Serotonin reuptake**
- D- Glutamate release

**11- Duloxetine is extensively metabolized in the:**

- A- Kidneys
- B- Spleen
- C- Liver**
- D- Stomach

**12- Levomilnacipran is metabolized primarily by:**

- A- CYP2D6
- B- CYP2C9
- C- CYP3A4**
- D- CYP1A2

**13- Which SNRI may increase blood pressure?**

- A- Levomilnacipran
- B- Duloxetine
- C- Venlafaxine
- D- All of the above**

**14- Duloxetine differ from venlafaxine in**

- A- May increase blood pressure
- B- Cause sexual dysfunction
- C- Inhibit serotonin and norepinephrine reuptake in all doses**
- D- Inducer of CYP2D6 isoenzymes

**15- Which is a tertiary amine TCA?**

- A- Desipramine
- B- Nortriptyline
- C- Imipramine**
- D- Protriptyline

**16- Is a TCA relatively selective inhibitor of norepinephrine reuptake**

- A- Imipramine
- B- Desipramine**
- C- Clomipramine
- D- Nortriptyline

**17- Blurred vision and xerostomia are adverse effect of TCAs because of blocked of**

- A- Alpha adrenergic receptor
- B- Histamine H1 receptors
- C- Muscarinic receptors**
- D- Serotonergic receptors

**18- Blockade of Alpha adrenergic receptors causing**

- A- Dry mouth
- B- Reflex bradycardia
- C- Orthostatic hypertension
- D- Orthostatic hypotension**

**19- TCAs must be used cautiously in bipolar disorder because they may:**

- A- Induce mania**
- B- Induce depressed state
- C- Induce hypertension
- D- Cause hypoglycemia

**20- MAOIs inhibit MAO in :**

- A- Lungs and brain
- B- Liver and brain**
- C- Brain only
- D- Gut only

**21- Cause hypertensive crisis with tyramine**

- A- SNRIs
- B- SSRIs
- C- MAOIs**
- D- TCAs

**22- Bupropion is useful for:**

- A- Insomnia
- B- Smoking cessation**
- C- Migraine
- D- Hypertension

**23- Bopropion is metabolized by**

- A- CYP2D6
- B- CYP2B6**
- C- CYP3A4
- D- CYP2A1

**24- Mirtazapine antagonist which receptor?**

- A- Histamine receptors
- B- Serotonin receptors
- C-  $\alpha_2$  receptors**
- D-  $\beta_2$  receptors

**25- Which antidepressant is most sedating?**

- A- SSRIs
- B- Mirtazapine**
- C- Bupropion
- D- Venlafaxine

**26- ..... associated with hepatotoxicity**

**.....associated with priapism**

- A- Nefazodone/ Trazodone**
- B- Nefazodone /Bupropion
- C- Trazodone/ vilazodone
- D- Trazodone /Nefazodone

**27- Vilazodone is a serotonin reuptake inhibitor and:**

- A-  $\alpha 1$  antagonist
- B-  $\beta$  blocker
- C- 5-HT<sub>1A</sub> partial agonist**
- D- Dopamine agonist

**28- Vortioxetine has actions on which receptors?**

- A- 5-HT<sub>2</sub> and 5-HT<sub>3</sub>
- B- 5-HT<sub>2A</sub> and 5-HT<sub>2C</sub>
- C- Alpha 2 only
- D- 5-HT<sub>7</sub>**

**29- Which class has the highest risk in overdose?**

- A- SSRIs
- B- SNRIs
- C- Atypicals
- D- TCAs**

**30- A 45-year-old man has major depressive disorder with chronic neuropathic pain from diabetic neuropathy. He previously tried an SSRI without improvement. Which drug class is most suitable for treating both depression and neuropathic pain?**

- A- SNRIs**
- B- SSRIs
- C- TCAs
- D- MAOIs

31- A 34-year-old man on an antidepressant develops severe headache, hypertension (210/120), sweating, and palpitations after eating aged cheese at a wedding.

**Which class of antidepressants caused this**

- A- SSRIs
- B- MAOIs**
- C- SNRIs
- D- TCAs

**32- Which drug has a very low incidence of sexual dysfunction**

- A- Fluoxetine
- B- Venlafaxine
- C- Bupropion**
- D- Duloxetine

**33- May cause sleep disturbances**

- A- Mirtazapine
- B- Nefazodone
- C- Duloxetine**
- D- Imipramine

**34- Mechanism of action of the group that used in depressed patient unresponsive to TCAs and SSRIs,**

Last line agent in treatment

- A- Inhibit reuptake of serotonin
- B- Increase concentration of the neurotransmitter in synaptic cleft
- C- Inhibit MAO enzyme**
- D- Increase degradation of neurotransmitter

**35- TCAs have strong antagonistic activity at which receptor to cause sedation?**

- A- Dopamine
- B- Muscarinic
- C- Histamine H<sub>1</sub>**
- D- NMDA

**36- TCAs used to prevent migraine headache and treat chronic pain syndromes, particularly which one of them**

- A- Amitriptylin
- B- Desipramine
- C- Imipramine
- D- Nortriptyline

**37- Symptoms of depression are included**

- A- Feeling of sadness and Hopelessness
- B- Inability to experience pleasure in usual activities
- C- Loss of energy and Suicidal thoughts
- D- All of the above

## Formatives

### & Previous Exams

**38- MAO enzyme is needed for inactivation of**

- A- GABA
- B- Glycin
- C- Catecholamines
- D- Acetyl choline

**39- Administration of MAOI would most likely to cause changes in the CNS concentration of**

- A- Ach
- B- Norepinephrine
- C- Histamine
- D- GABA

**40- Most of antidepressants produce their effects**

- A- Soon after administration
- B- A day later
- C- After 3-4 weeks
- D- After 6 months

**41- Priapism is a possible side effect of**

- A- Amitriptyline
- B- Trazodone
- C- Mirtazapine
- D- Bupropion

**42- Which of the following antidepressants may be used in smoking cessation**

- A- Mirtazapine
- B- Duloxetine
- C- Phenelzine
- D- Bupropion

**43- What is the primary mechanism of action of tricyclic antidepressants (TCAs)?**

- A- Increasing dopamine levels
- B- Blocking serotonin and norepinephrine reuptake**
- C- Inhibiting monoamine oxidase enzymes
- D- Enhancing GABAergic activity

**44- What is the common side effect of SSRI antidepressants**

- A- Hypertension
- B- Sexual dysfunction**
- C- Increased appetite
- D- Hypoglycemia

**45- Regarding the antidepressant imipramine**

- A- It acts by inhibiting reuptake of serotonin and norepinephrine**
- B- It is selective serotonin reuptake inhibitor
- C- inhibits monoamine oxidase enzyme
- D- Dose adjustment is only required in presence of renal diseases.

**46- Atypical antidepressants include**

- A- Sertraline
- B- Fluoxetine
- C- Bupropion**
- D- Selegiline

**47- Which of the following is most likely to cause orthostatic hypotension**

- A- Desipramine**
- B- Venlafaxine
- C- Fluoxetine
- D- Sertraline

**48- TCAs used to prevent migraine headache and treat chronic pain syndromes, particularly which one of them**

- A- Amitriptylin**
- B- Desipramine
- C- Imipramine
- D- Nortriptyline

**49- Which of the following is**

- a selective serotonin reuptake inhibitor (SSRI)
- A- Amitriptyline
- B- Fluoxetine**
- C- Venlafaxine
- D- Bupropion

**50- Which antidepressant is most commonly associated with the side effect of hypertension (especially at higher doses)**

- A- Sertraline
- B- Mirtazapine
- C- Venlafaxine**
- D- Citalopram

# Local Anesthetics

**1- Local anesthetics produce their effect mainly by blocking which ion channel.....?**

- A- Chloride
- B- Calcium
- C- Sodium**
- D- Potassium

**2- Local anesthetics block pain sensation by preventing.....?**

- A- Neurotransmitter release
- B- Repolarization of the nerve membrane
- C- Depolarization of all cell membranes
- D- Action potential conduction**

**3- Which of the following isn't a characteristic of an ideal local anesthetic.....?**

- A- Minimal systemic toxicity
- B- Permanent nerve blockade**
- C- Long duration
- D- Rapid onset

**4- The lipophilic part of a local anesthetic molecule is.....?**

- A- Amide linkage
- B- Ester linkage
- C- Aromatic ring**
- D- Ionizable amino group

**5- Which component of local anesthetics is responsible for membrane penetration.....?**

- A- Amide group
- B- Ester group
- C- Aromatic ring**
- D- Ester linkage

**6- Which component of local anesthetics is responsible for Na<sup>+</sup> channel blockade.....?**

- A- Ionizable amino group**
- B- Amide group
- C- Ester group
- D- Aromatic ring

**7- Ester-type local anesthetics are mainly metabolized by.....?**

- A- CNS
- B- Lungs
- C- Plasma esterases**
- D- Plasma

**8- Amide-type local anesthetics are mainly metabolized in the.....?**

- A- Heart
- B- Liver**
- C- Kidney
- D- Plasma

**9- Local anesthetic activity is increased at.....?**

- A- Very low pH
- B- Alkaline pH**
- C- Neutral pH
- D- Acidic pH

**10- Inflamed tissues are resistant to local anesthetics because they are.....?**

- A- Edematous
- B- Ischemic
- C- Acidic**
- D- Alkaline

**11- The ionized form of local anesthetics.....?**

- A- Has higher activity
- B- Cannot penetrate nerve membranes**
- C- Is lipid soluble
- D- Easily crosses membranes

**12- Local anesthetics prevent nerve conduction by.....?**

- A- Stimulating Ca<sup>2+</sup> influx
- B- Increasing K<sup>+</sup> permeability
- C- Blocking Na<sup>+</sup> permeability**
- D- Depolarizing the nerve

**13- Which nerve fibers are most sensitive to local anesthetics.....?**

- A- Large non myelinated fibers
- B- Motor fibers
- C- Small myelinated fibers**
- D- Large myelinated fibers

**14- The first sensation lost after local anesthesia is.....?**

- A- Temperature
- B- Pain**
- C- Pressure
- D- Touch

**15- Motor fibers are relatively.....?**

- A- Permanently blocked
- B- Resistant**
- C- Equally sensitive
- D- More sensitive

**16- Potency of local anesthetics is directly related to.....?**

- A- Renal clearance
- B- Ionization
- C- Lipid solubility**
- D- Water solubility

**17- Which local anesthetic is more lipid soluble and more potent.....?**

- A- Mepivacaine
- B- Tetracaine**
- C- Procaine
- D- Lidocaine

**18- Vasoconstrictors prolong local anesthetic action by.....?**

- A- Blocking Na<sup>+</sup> channels
- B- Decreasing absorption**
- C- Increasing absorption
- D- Increasing metabolism

**19- The most commonly used vasoconstrictor with local anesthetics is.....?**

- A- Epinephrine**
- B- Dopamine
- C- Propranolol
- D- Acetyl choline

**20- Cocaine causes vasoconstriction because it.....?**

- A- Stimulates beta receptors
- B- Inhibits catecholamine reuptake**
- C- Blocks Ca<sup>2+</sup> channels
- D- Blocks Na<sup>+</sup> channels

**21- Which local anesthetic doesn't need a vasoconstrictor.....?**

- A- Benzocaine
- B- Bupivacaine**
- C- Procaine
- D- Lidocaine

**22- All local anesthetics cause vasodilation except.....?**

- A- Cocaine**
- B- Bupivacaine
- C- Procaine
- D- Lidocaine

**23- CNS toxicity of local anesthetics initially causes.....?**

- A- Paralysis
- B- Coma
- C- CNS stimulation**
- D- Respiratory depression

**24- Death from local anesthetic toxicity usually occurs due to.....?**

- A- Respiratory failure**
- B- Hepatic failure
- C- Renal failure
- D- Cardiac arrest

**25- Bupivacaine is associated with severe.....?**

- A- Renal toxicity
- B- Bronchospasm
- C- Arrhythmias**
- D- Hypotension only

**26- Hypersensitivity reactions are more common with.....?**

- A- Lidocaine only
- B- Tetracaine only
- C- Ester anesthetics**
- D- Amide anesthetics

**27- Benzocaine is unsuitable for injection because it.....?**

- A- Is rapidly metabolized
- B- Is too potent
- C- Lacks tertiary amino group**
- D- Is highly toxic

**28- Surface anesthesia is commonly used for.....?**

- A- Joint anesthesia
- B- Mucous membranes**
- C- Epidural anesthesia
- D- Deep surgical procedures

**29- Ring block anesthesia is a type of.....?**

- A- Epidural anesthesia
- B- Spinal anesthesia
- C- Infiltration anesthesia**
- D- Surface anesthesia

**30- Spinal anesthesia is injected into the.....?**

- A- Ventricular system
- B- Subarachnoid space**
- C- Epiarachnoid space
- D- Epidural space

**31- Spinal anesthesia is usually given between.....?**

- A- S1–S5
- B- C1–C5
- C- L2–L5**
- D- T1–T5

**32- A common complication of spinal anesthesia is.....?**

- A- Hyperthermia
- B- Hypotension**
- C- Bradycardia
- D- Hypertension

**33- Lidocaine belongs to which class of antiarrhythmics.....?**

- A- Class III
- B- Class II
- C- Class IB**
- D- Class IA

**34- Lidocaine is effective in treating.....?**

- A- Atrial fibrillation
- B- Supraventricular tachycardia
- C- Ventricular arrhythmias**
- D- AV block

**35- Lidocaine is metabolized mainly in the....?**

- A- Lung
- B- Liver**
- C- Plasma
- D- Kidney

**36- A black box warning for lidocaine concerns its use in.....?**

- A- Cardiac patients
- B- Children with teething pain**
- C- Pregnant women
- D- Elderly patients

**37- Bupivacaine is.....?**

- A- Used topically only
- B- Long acting**
- C- Less potent than lidocaine
- D- Short acting

**38- Ropivacaine has duration of action....?**

- A- Short
- B- Medium
- C- Similar to bupivacaine**
- D- Less than lidocaine

**39- Mepivacaine is contraindicated in.....?**

- A- Elderly
- B- Neonates**
- C- Children
- D- Adults

**40- Prilocaine may cause.....?**

- A- Arrhythmia
- B- Nephrotoxicity
- C- Methemoglobinemia**
- D- Hepatotoxicity

**41- Tetracaine is characterized by....?**

- A- Very rapid metabolism
- B- Medium onset and duration
- C- Slow onset and long duration**
- D- Rapid onset and short duration

**42- Tetracaine is commonly used in.....?**

- A- Cardiac arrhythmias
- B- Spinal anesthesia**
- C- Topical anesthesia
- D- All of above

**43- Cocaine is classified as.....?**

- A- Opioid
- B- Natural local anesthetic**
- C- Synthetic ester
- D- Synthetic amide

**44- Which local anesthetic can cause addiction.....?**

- A- Benzocaine
- B- Cocaine**
- C- Procaine
- D- Lidocaine

**45- Local anesthetics depress smooth muscle of.....?**

- A- Uterus only
- B- Bronchi only
- C- GI tract only
- D- All of the above**

**46- Neuromuscular blockade by local anesthetics occurs at.....?**

- A- Topical use only
- B- Large doses**
- C- Therapeutic doses
- D- Low doses

**47- Plasma levels of lidocaine are increased in patients with.....?**

- A- Hypertension
- B- Diabetes
- C- CHF**
- D- Renal failure

**48- Which anesthetic is highly toxic to neonates.....?**

- A- Tetracaine
- B- Mepivacaine**
- C- Procaine
- D- Lidocaine

**49- Benzocaine is an ester of.....?**

- A- Acetic acid
- B- Salicylic acid
- C- Benzoic acid
- D- PABA**

**50- Which anesthetic is available only parenterally.....?**

- A- Benzocaine
- B- Bupivacaine**
- C- Cocaine
- D- Lidocaine

**51- Local anesthetics stabilize the membrane by blocking.....?**

- A- Hyperpolarization
- B- Depolarization**
- C- Mg+2 entry

**52- Phenylephrine concentration used as vasoconstrictor is.....?**

- A- 1%
- B- 5%
- C- 0.05%**
- D- 0.5%

**53- Sympathomimetic amines may cause local tissue damage due to.....?**

- A- Infection
- B- Hypoxia**
- C- Hyperoxia

**54- Absorption of local anesthetics is fastest from.....?**

- A- Pharynx
- B- Urinary bladder
- C- Trachea**

**55- Local anesthetics are usually ineffective on.....?**

- A- Small nerves
- B- Inflamed tissue**
- C- Mucous membranes
- D- S.C

**56- Which anesthetic isn't suitable for injection.....?**

- A- Benzocaine**
- B- Bupivacaine
- C- Lidocaine
- D- Ropivacaine

**57- Local anesthetic-induced CNS excitation is due to block of...?**

- A- Sensory neurons
- B- Motor neurons
- C- Inhibitory neurons**
- D- Excitatory neurons

**58- Which anesthetic has medium onset and duration and causes methemoglobinemia.....?**

- A- Ropivacaine
- B- Procaine
- C- Prilocaine**
- D- Lidocaine

## فورماتيف وامتحانات سابقة

**59- Bupivacain is.....?**

- A- Slow onset, Long duration
- B- Rapid onset, Slow duration
- C- Rapid onset, Rapid duration
- D- Slow onset, Slow duration

**60- What effect does inflammation in tissues have on the activity of local anesthetics.....?**

- A- Causes prolonged duration of action
- B- Has no significant impact
- C- Reduces the activity due to acidic pH
- D- Increases the activity due to enhanced absorption

**61- Which local anesthetic is contraindicated in obstetrical anesthesia due to its high toxicity to neonates.....?**

- A- Bupivacaine
- B- Mepivacaine
- C- Tetracaine
- D- Prilocaine

**62- Choose the correct sentence of the following.....?**

- A- Bupivacaine may produce serious arrhythmias and tachycardia
- B- Inflamed tissues are somewhat sensitive to local anesthetics agents
- C- Elimination of the amide groups by plasma esterases
- D- Local anesthetics are acidic compounds, and their activities are strongly PH dependent

**63- Adam has a history of allergy to ester type of local anesthetics, an acceptable choice for producing nerve block anesthesia will be.....?**

- A- Tetracaine
- B- Procaine
- C- Mepivacaine
- D- Chloroprocaine

**64- Example of local anesthetic with intrinsic sympathomimetic vasoconstrictive properties.....?**

- A- Lidocaine
- B- Procaine
- C- Cocaine
- D- Mepivacaine

**65- Inclusion of a vasoconstrictor in the local anesthetic solution is recommended to.....?**

- A- Decrease the duration of action
- B- Bloodless surgical field
- C- Increase the systemic absorption.
- D- Terminate its action.

**66- One of the anesthetic drugs is commonly used topically & parenterally and may be used as antiarrhythmic.....?**

- A- Cocaine
- B- Mepivacaine
- C- Lidocaine
- D- Bupivacaine

**67- Amal has a history of allergy to ester type of local anesthetics, an acceptable choice for producing nerve block anesthesia will be.....?**

- A- Procaine
- B- Tetracaine
- C- Mepivacaine
- D- Chloroprocaine